# EPR pre-generation on ten benchmarks

ODG, CGP, ACGP and compiler-driven Dynamic, memories per core $K = 1$ to $\infty$

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# Chapter 1

# Setup

## 1.1 Architecture and workloads

- $4 \times 4$ mesh of 16 cores; each core has 16 data slots, 14 occupied at the start and 2 kept free for migration, so circuits have $16 \times 14 = 224$ logical qubits.
- Benchmarks: ten MQT Bench 2.3.0 circuits at 224 qubits (`benchmarks/`, verified by `benchmarks/verify.py`). Each is mapped once (T-HQA placer, OEE initial placement, mapping seed 42) and routed with XY and with history routing (history counts the route visits of all preceding layers, i.e. cumulative history, routing capacity 8); every strategy and every $K$ sees the same frozen trace.
- Physics (`config/final_config/grid_4x4_ace_3.json`): memory efficiency 0.8, memory fidelity 0.95, coherence 1 s, detector efficiency 0.95. The classical delay is set by the topology code to about 0.1 ms between neighbours.

## 1.2 Strategies and policy

ODG generates every pair on demand. CGP and ACGP generate pairs continuously into a cache. Dynamic is compiler driven: the planner chooses, before the run, the layer in which each pair is started. Everything except $K$ and the memory split is the default policy of the code (branch `history-algo-update`, `ace/compiler/settings.py`):

| Setting | Value |
|---|---|
| Layer basis, lead, lookahead, age limit | active layers; 1, 2, 10 layers |
| Generation budget | 8 launches per core per active layer |
| Preparation priority | trace order (`trace`) |
| Demand guard / refill on release | on / on |
| Admission | per-qubit, event driven, ready pairs first, longer chains first |
| Empty-layer gap | 10 ns |
| Compiler reservation, pair use | 50 ms, handoff |
| Physics seeds | 60 and 61 |

**Table 1.1:** Default policy used in every run.

## 1.3 Memories per core and the split

| $K$ (memories per core) | 1 | 2 | 3 | 4 | 6 | 8 | $\infty$ (64) |
|---|---|---|---|---|---|---|---|
| Dynamic may use | 1 | 2 | 3 | 4 | 6 | 6 | 64 |
| CGP / ACGP cache ($\lfloor 0.75K \rfloor$) | 0 | 1 | 2 | 3 | 4 | 6 | 48 |

**Table 1.2:** Pre-generation share of the $K$ memories. At $K = 8$ six memories are reserved for pre-generation. CGP/ACGP use $\lfloor 0.75K \rfloor$ because caching in all $K$ memories starves requests (smoke test: 150–1060 ms per hop at $K = 1$–4); at $K = 1$ they therefore equal ODG.

## 1.4 Metrics

A *hop* is one request between neighbouring cores; a *chain* is all hops of one qubit in one layer.

**Latency** mean time per hop from eligibility to EPR delivery, including any wait for a free memory.

**Fidelity** mean fidelity of the delivered pairs (not a circuit fidelity).

**Coverage** share of hops that found a pair already made: generation phase under 0.35 ms, i.e. only the reservation handshake. Same definition for every strategy. Chain full coverage: every hop of the chain ready.

**Utilisation** pairs used divided by pairs generated. **Wastage** = 100− utilisation: pairs that expired or were never used. ODG makes no pairs in advance, so both are not applicable to it.

Bars are the mean of the two seeds; whiskers span the two seeds.

# Chapter 2

# Results

## 2.1 QFT, XY routing

11076 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.1.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.782 | 4.60 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.970 | 3.25 | 0.9424 | 77.9 | 60.1 | 4.24 | 95.8 |
| ACGP | 0.976 | 3.30 | 0.9421 | 77.6 | 60.0 | 4.39 | 95.6 |
| Dynamic (8) | 0.747 | 2.25 | 0.9439 | 76.8 | 64.5 | 1.23 | 98.8 |
| Dynamic ($\infty$) | 0.616 | 1.65 | 0.9442 | 84.0 | 71.9 | 1.18 | 98.8 |

**Table 2.1:** QFT, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

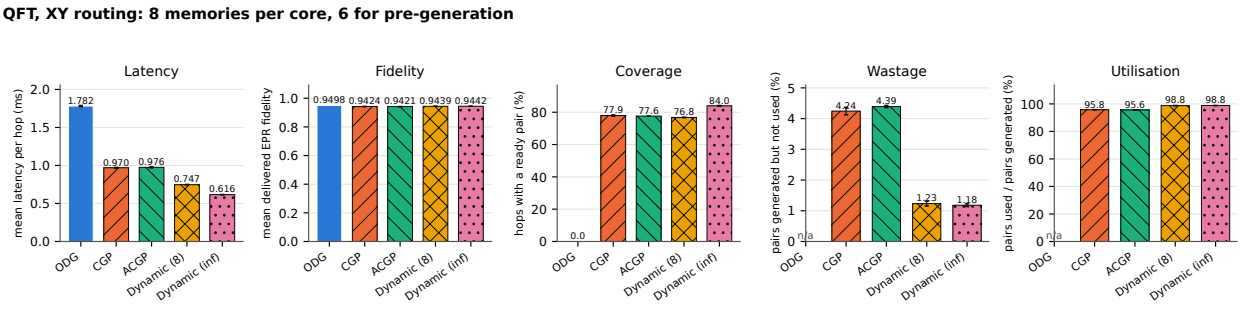



**Figure 2.1:** QFT, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.1.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.887 | 4.80 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 2.331 | 8.95 | 0.9401 | 63.0 | 41.7 | 5.87 | 94.1 |
| ACGP | 2.325 | 8.95 | 0.9402 | 63.4 | 43.9 | 5.76 | 94.2 |
| Dynamic (4) | 1.295 | 4.70 | 0.9425 | 63.7 | 54.1 | 3.89 | 96.1 |
| Dynamic ($\infty$) | 0.616 | 1.65 | 0.9442 | 84.0 | 71.9 | 1.18 | 98.8 |

**Table 2.2:** QFT, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

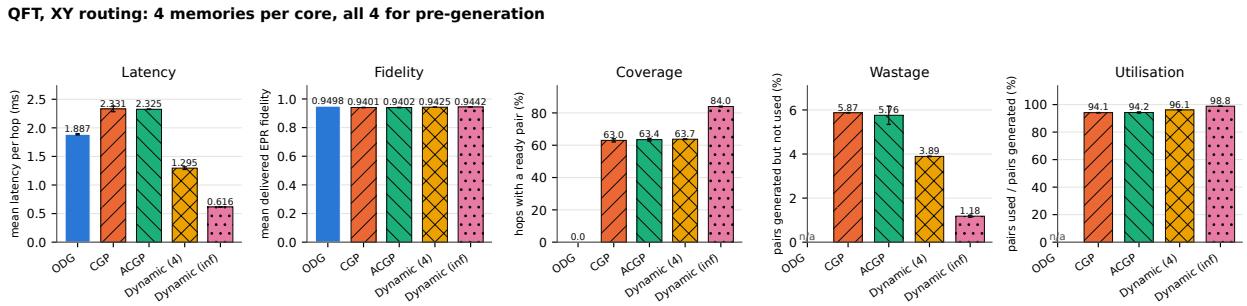



**Figure 2.2:** QFT, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.1.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 6.275 | 2.816 | 2.109 | 1.887 | 1.788 | 1.782 | 1.782 |
| CGP | 6.275 | 4.177 | 2.885 | 2.331 | 1.237 | 0.970 | 0.457 |
| ACGP | 6.275 | 4.118 | 2.974 | 2.325 | 1.247 | 0.976 | 0.458 |
| Dynamic | 9.041 | 3.528 | 1.877 | 1.295 | 0.866 | 0.747 | 0.616 |

**Table 2.3:** QFT, XY routing: latency (mean latency per hop (ms)) for every K.

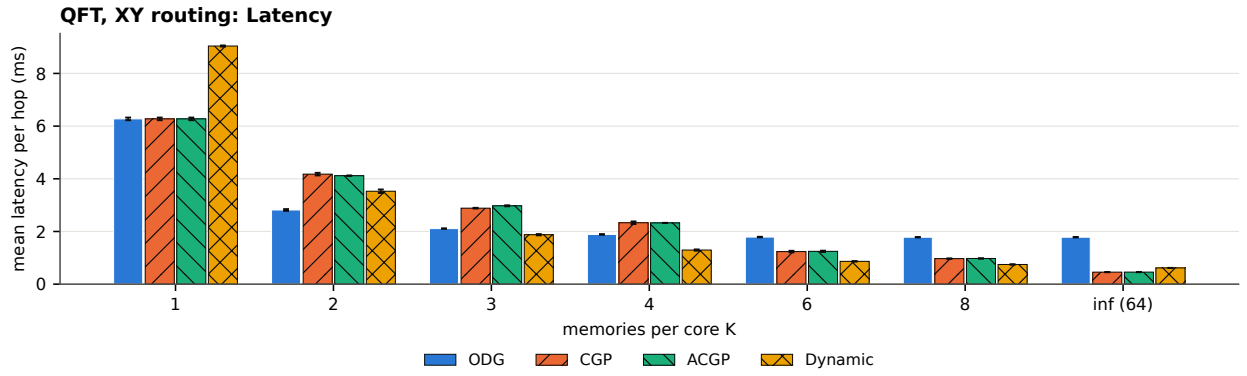



**Figure 2.3:** QFT, XY routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9441 | 0.9415 | 0.9401 | 0.9427 | 0.9424 | 0.9422 |
| ACGP | 0.9498 | 0.9441 | 0.9414 | 0.9402 | 0.9425 | 0.9421 | 0.9422 |
| Dynamic | 0.9486 | 0.9440 | 0.9429 | 0.9425 | 0.9432 | 0.9439 | 0.9442 |

**Table 2.4:** QFT, XY routing: fidelity (mean delivered EPR fidelity) for every K.

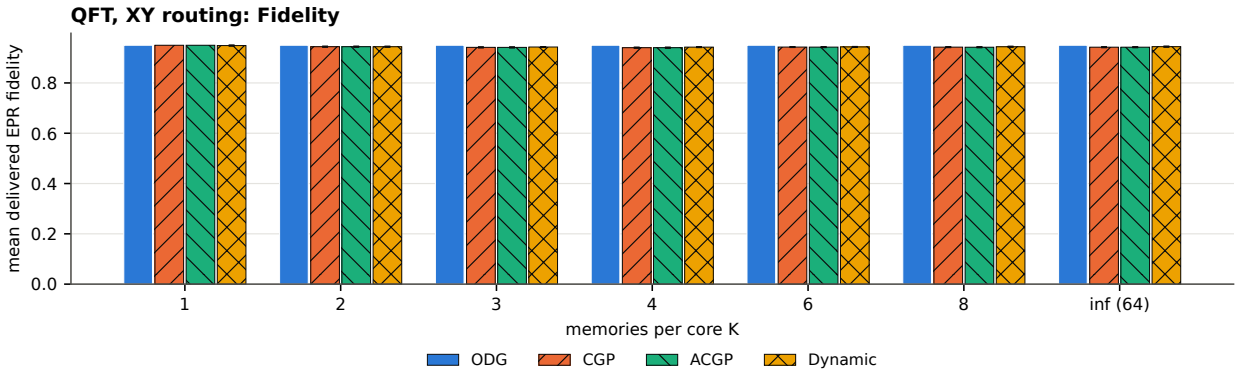



**Figure 2.4:** QFT, XY routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 26.8 | 50.5 | 63.0 | 64.2 | 77.9 | 99.3 |
| ACGP | 0.0 | 27.0 | 49.0 | 63.4 | 63.9 | 77.6 | 99.1 |
| Dynamic | 8.9 | 36.3 | 52.4 | 63.7 | 76.8 | 76.8 | 84.0 |

**Table 2.5:** QFT, XY routing: coverage (hops with a ready pair (%)) for every K.

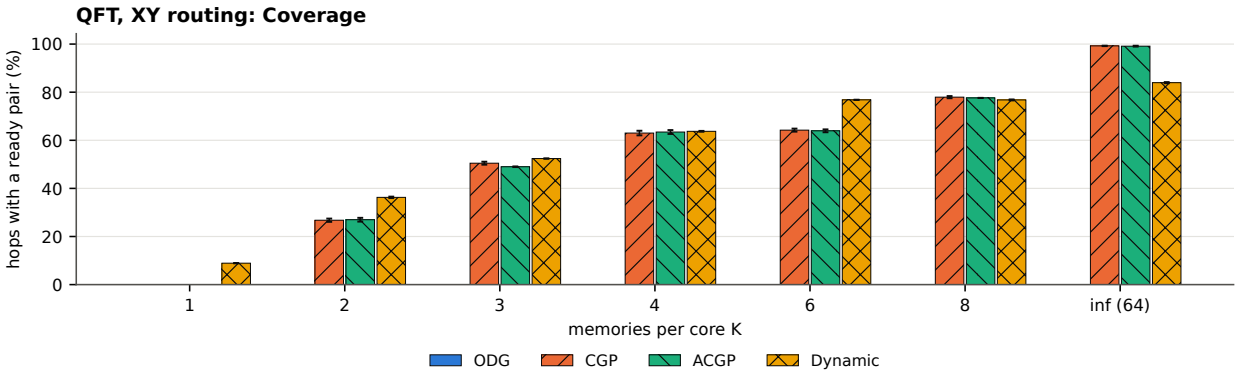



**Figure 2.5:** QFT, XY routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 7.08 | 5.66 | 5.87 | 4.34 | 4.24 | 13.73 |
| ACGP | – | 7.04 | 6.13 | 5.76 | 4.49 | 4.39 | 13.73 |
| Dynamic | 60.80 | 14.59 | 6.08 | 3.89 | 2.05 | 1.23 | 1.18 |

**Table 2.6:** QFT, XY routing: wastage (pairs generated but not used (%)) for every K.

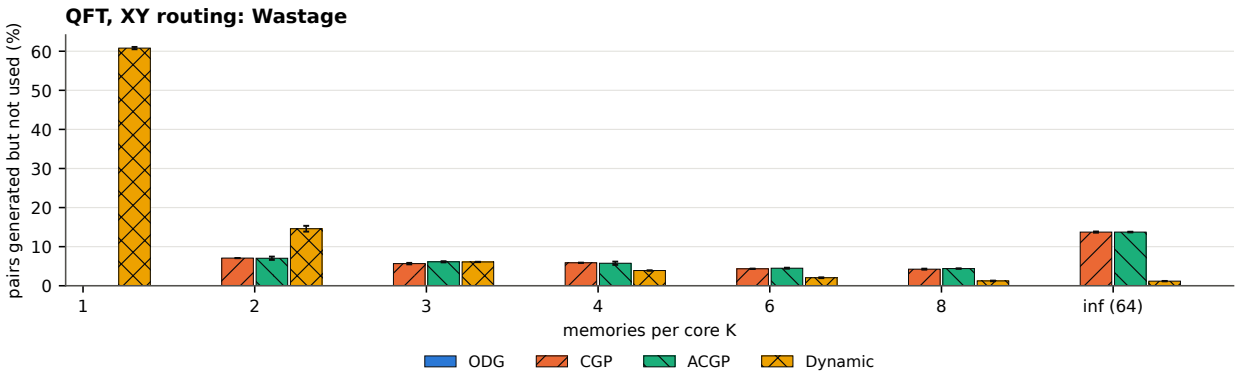



**Figure 2.6:** QFT, XY routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 92.9 | 94.3 | 94.1 | 95.7 | 95.8 | 86.3 |
| ACGP | – | 93.0 | 93.9 | 94.2 | 95.5 | 95.6 | 86.3 |
| Dynamic | 39.2 | 85.4 | 93.9 | 96.1 | 98.0 | 98.8 | 98.8 |

**Table 2.7:** QFT, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

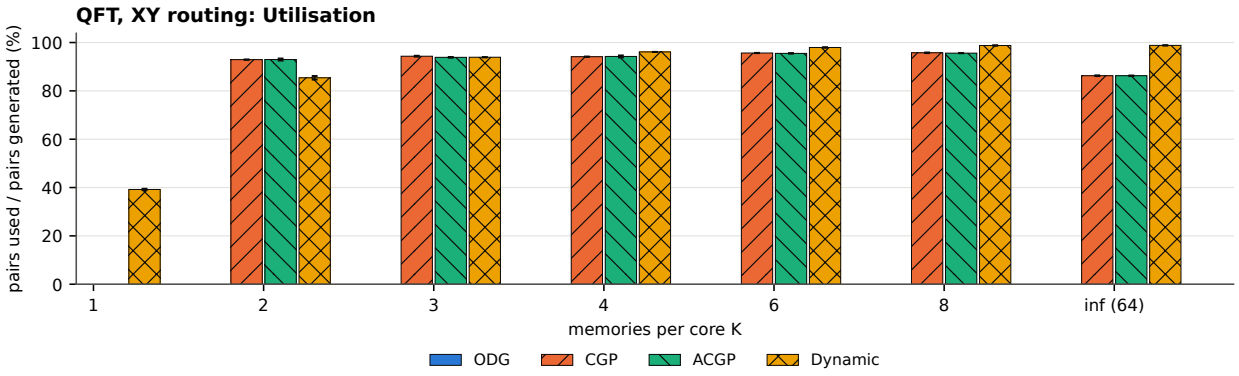



**Figure 2.7:** QFT, XY routing: utilisation for every K.

## 2.2 QFT, history routing

11076 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.2.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.773 | 4.40 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.966 | 3.10 | 0.9422 | 73.8 | 52.6 | 4.97 | 95.0 |
| ACGP | 0.917 | 2.95 | 0.9425 | 76.6 | 57.4 | 4.34 | 95.7 |
| Dynamic (8) | 0.693 | 2.00 | 0.9442 | 79.6 | 65.7 | 1.40 | 98.6 |
| Dynamic ($\infty$) | 0.584 | 1.45 | 0.9450 | 85.6 | 72.4 | 1.46 | 98.5 |

**Table 2.8:** QFT, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

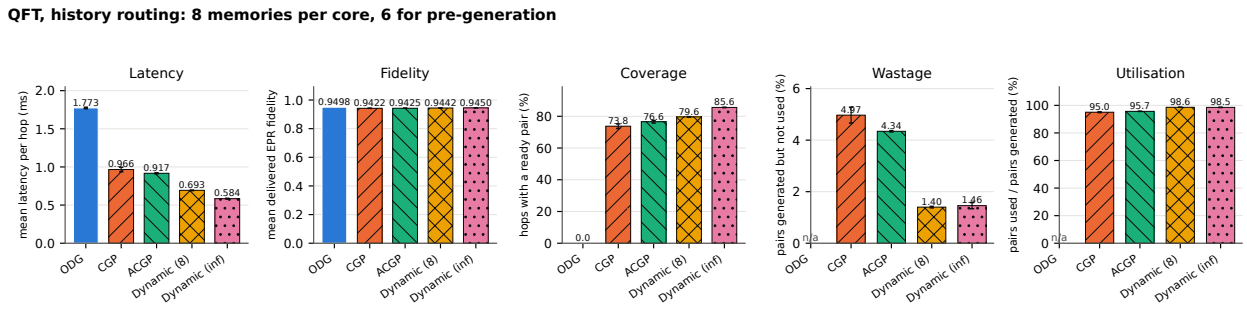



**Figure 2.8:** QFT, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.2.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.824 | 4.60 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 2.192 | 8.10 | 0.9413 | 63.5 | 40.3 | 5.37 | 94.6 |
| ACGP | 2.194 | 8.14 | 0.9412 | 62.3 | 38.3 | 5.44 | 94.6 |
| Dynamic (4) | 1.199 | 4.20 | 0.9423 | 65.8 | 55.5 | 4.45 | 95.6 |
| Dynamic ($\infty$) | 0.584 | 1.45 | 0.9450 | 85.6 | 72.4 | 1.46 | 98.5 |

**Table 2.9:** QFT, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

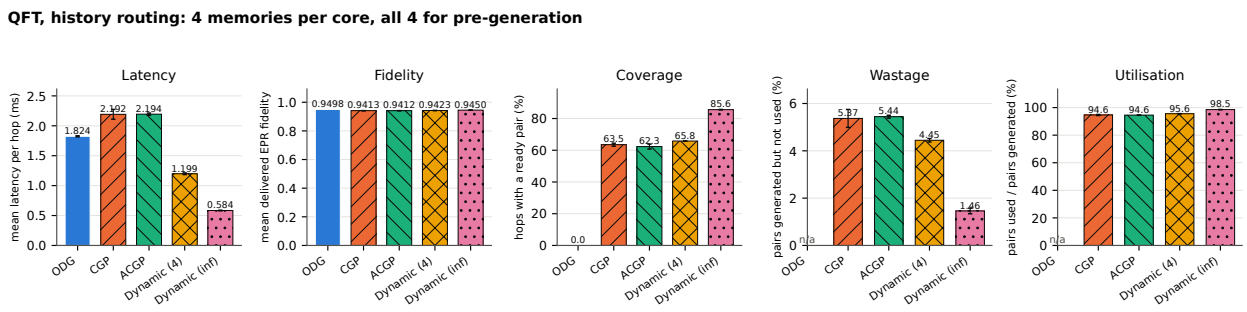



**Figure 2.9:** QFT, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.2.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 5.820 | 2.554 | 1.963 | 1.824 | 1.776 | 1.773 | 1.773 |
| CGP | 5.820 | 3.739 | 2.767 | 2.192 | 1.112 | 0.966 | 0.457 |
| ACGP | 5.820 | 3.748 | 2.696 | 2.194 | 1.113 | 0.917 | 0.455 |
| Dynamic | 8.812 | 3.153 | 1.749 | 1.199 | 0.797 | 0.693 | 0.584 |

**Table 2.10:** QFT, history routing: latency (mean latency per hop (ms)) for every K.

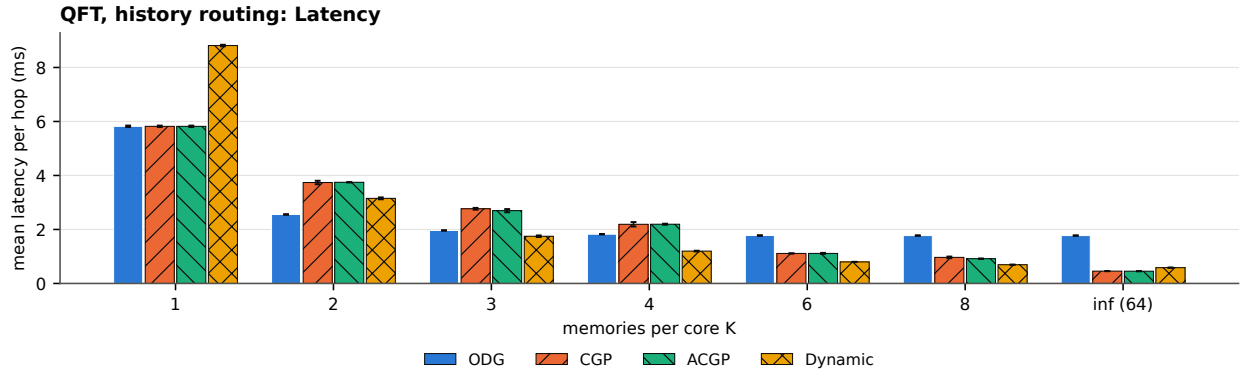



**Figure 2.10:** QFT, history routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9452 | 0.9427 | 0.9413 | 0.9433 | 0.9422 | 0.9417 |
| ACGP | 0.9498 | 0.9449 | 0.9426 | 0.9412 | 0.9433 | 0.9425 | 0.9418 |
| Dynamic | 0.9487 | 0.9442 | 0.9427 | 0.9423 | 0.9439 | 0.9442 | 0.9450 |

**Table 2.11:** QFT, history routing: fidelity (mean delivered EPR fidelity) for every K.

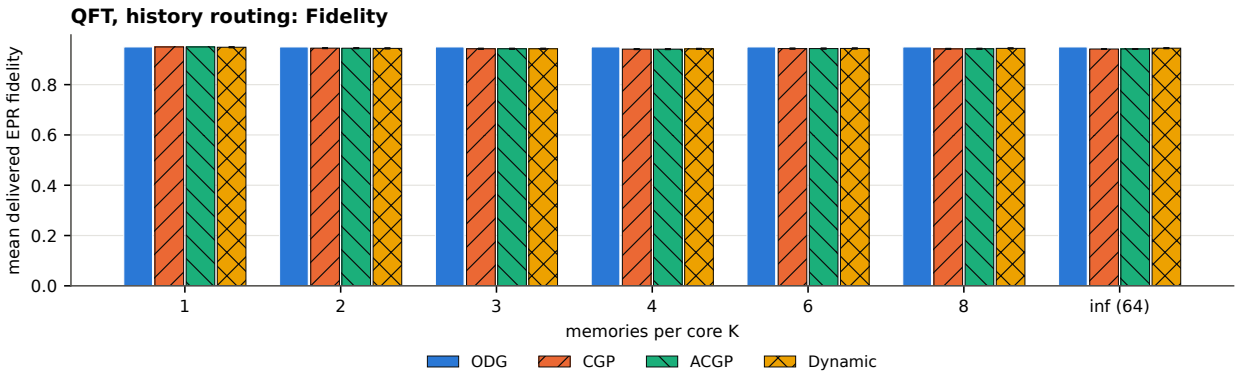



**Figure 2.11:** QFT, history routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 27.2 | 47.8 | 63.5 | 64.2 | 73.8 | 99.2 |
| ACGP | 0.0 | 28.2 | 49.7 | 62.3 | 65.3 | 76.6 | 99.5 |
| Dynamic | 8.4 | 36.4 | 53.8 | 65.8 | 79.2 | 79.6 | 85.6 |

**Table 2.12:** QFT, history routing: coverage (hops with a ready pair (%)) for every K.

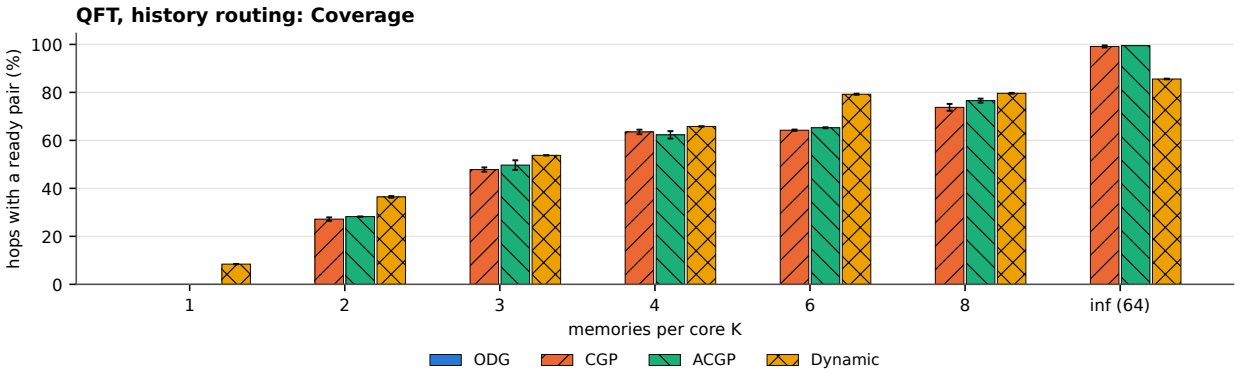



**Figure 2.12:** QFT, history routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 6.23 | 5.85 | 5.37 | 3.96 | 4.97 | 14.14 |
| ACGP | – | 6.07 | 5.58 | 5.44 | 3.74 | 4.34 | 13.66 |
| Dynamic | 63.26 | 14.98 | 6.07 | 4.45 | 2.25 | 1.40 | 1.46 |

**Table 2.13:** QFT, history routing: wastage (pairs generated but not used (%)) for every K.

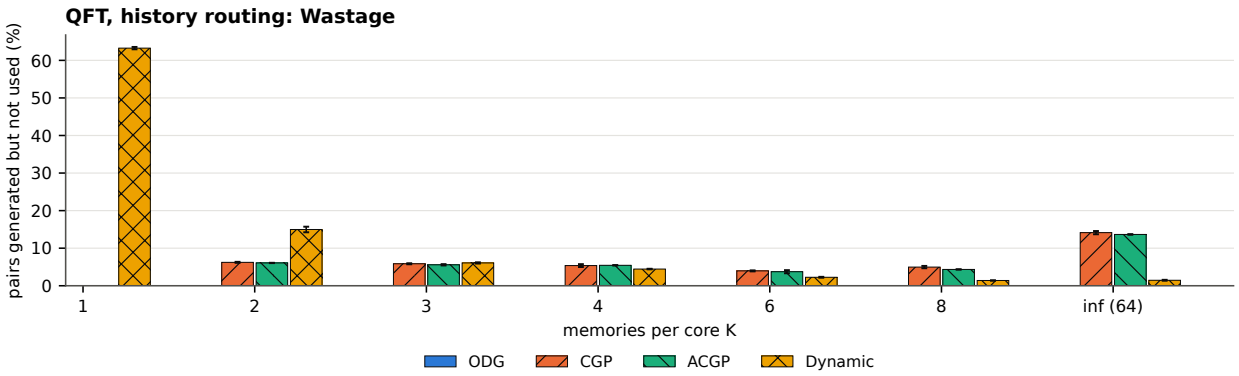



**Figure 2.13:** QFT, history routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 93.8 | 94.1 | 94.6 | 96.0 | 95.0 | 85.9 |
| ACGP | – | 93.9 | 94.4 | 94.6 | 96.3 | 95.7 | 86.3 |
| Dynamic | 36.7 | 85.0 | 93.9 | 95.6 | 97.7 | 98.6 | 98.5 |

**Table 2.14:** QFT, history routing: utilisation (pairs used / pairs generated (%)) for every K.

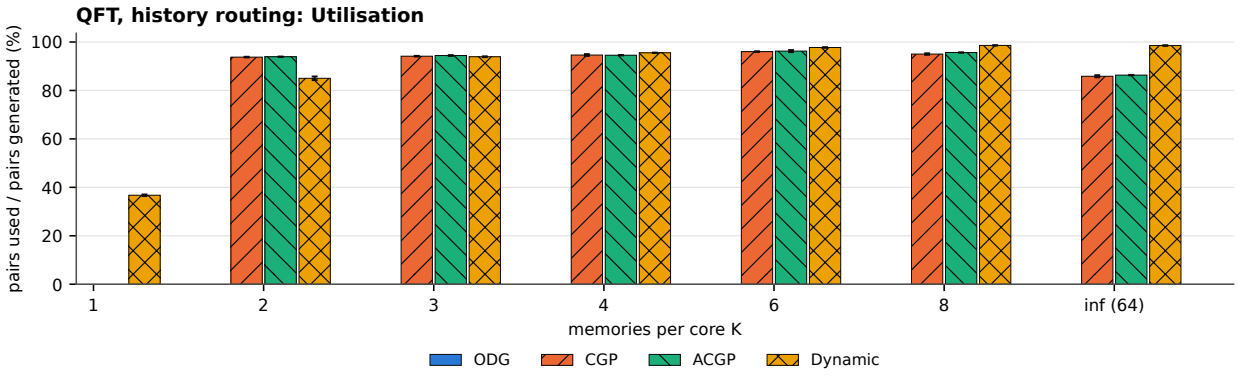



**Figure 2.14:** QFT, history routing: utilisation for every K.

## 2.3 BV, XY routing

23 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.3.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.550 | 3.40 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.648 | 1.60 | 0.9447 | 73.9 | 60.7 | 72.36 | 27.6 |
| ACGP | 0.648 | 1.60 | 0.9447 | 73.9 | 60.7 | 72.36 | 27.6 |
| Dynamic (8) | 0.691 | 1.70 | 0.9484 | 65.2 | 53.6 | 5.56 | 94.4 |
| Dynamic ($\infty$) | 0.691 | 1.70 | 0.9484 | 65.2 | 53.6 | 5.56 | 94.4 |

**Table 2.15:** BV, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

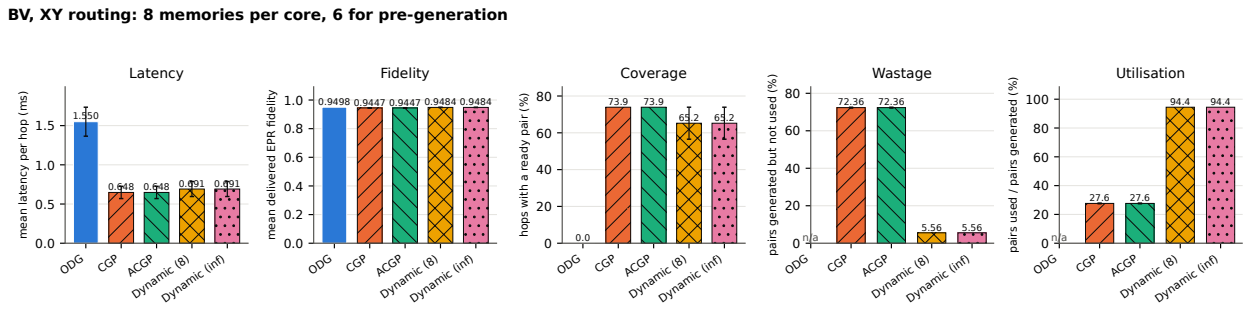



**Figure 2.15:** BV, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.3.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.550 | 3.40 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.811 | 2.40 | 0.9410 | 67.4 | 46.4 | 56.67 | 43.3 |
| ACGP | 0.811 | 2.40 | 0.9410 | 67.4 | 46.4 | 56.67 | 43.3 |
| Dynamic (4) | 0.691 | 1.70 | 0.9484 | 65.2 | 53.6 | 5.56 | 94.4 |
| Dynamic ($\infty$) | 0.691 | 1.70 | 0.9484 | 65.2 | 53.6 | 5.56 | 94.4 |

**Table 2.16:** BV, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

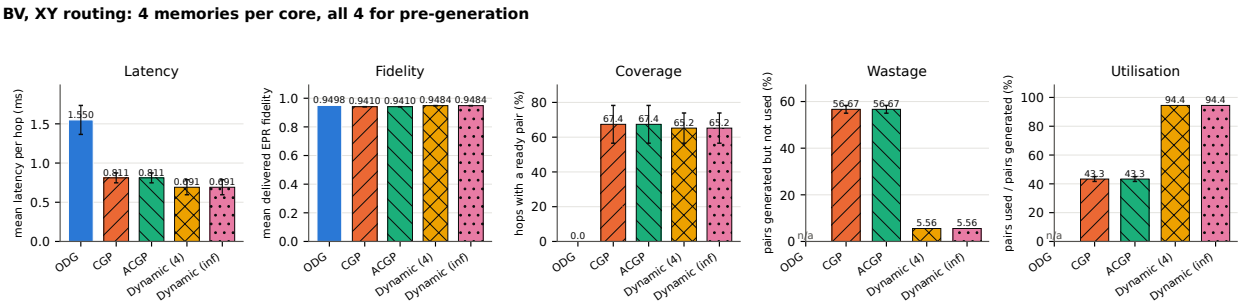



**Figure 2.16:** BV, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.3.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 1.550 | 1.550 | 1.550 | 1.550 | 1.550 | 1.550 | 1.550 |
| CGP | 1.550 | 1.228 | 0.948 | 0.811 | 0.526 | 0.648 | 0.504 |
| ACGP | 1.550 | 1.228 | 0.948 | 0.811 | 0.526 | 0.648 | 0.504 |
| Dynamic | 2.065 | 1.065 | 0.691 | 0.691 | 0.691 | 0.691 | 0.691 |

**Table 2.17:** BV, XY routing: latency (mean latency per hop (ms)) for every K.

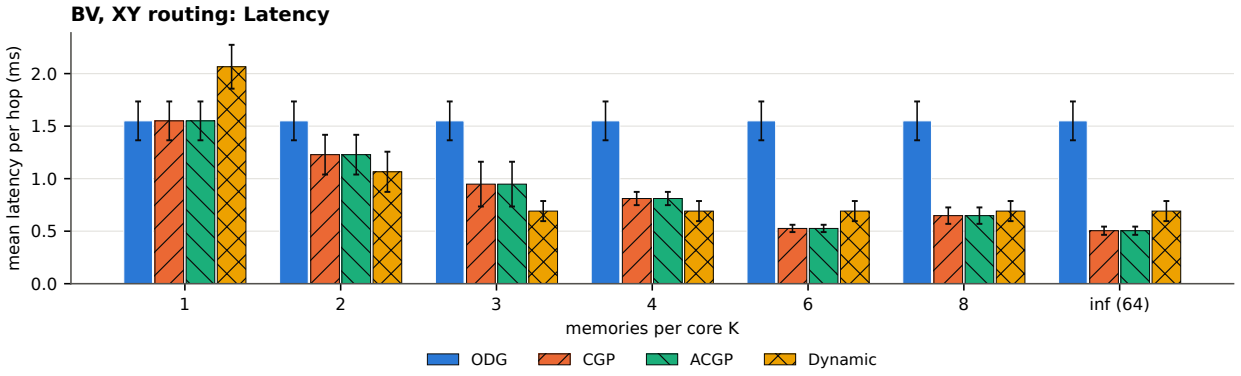



**Figure 2.17:** BV, XY routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9460 | 0.9432 | 0.9410 | 0.9434 | 0.9447 | 0.9473 |
| ACGP | 0.9498 | 0.9460 | 0.9432 | 0.9410 | 0.9434 | 0.9447 | 0.9473 |
| Dynamic | 0.9493 | 0.9477 | 0.9484 | 0.9484 | 0.9484 | 0.9484 | 0.9484 |

**Table 2.18:** BV, XY routing: fidelity (mean delivered EPR fidelity) for every K.

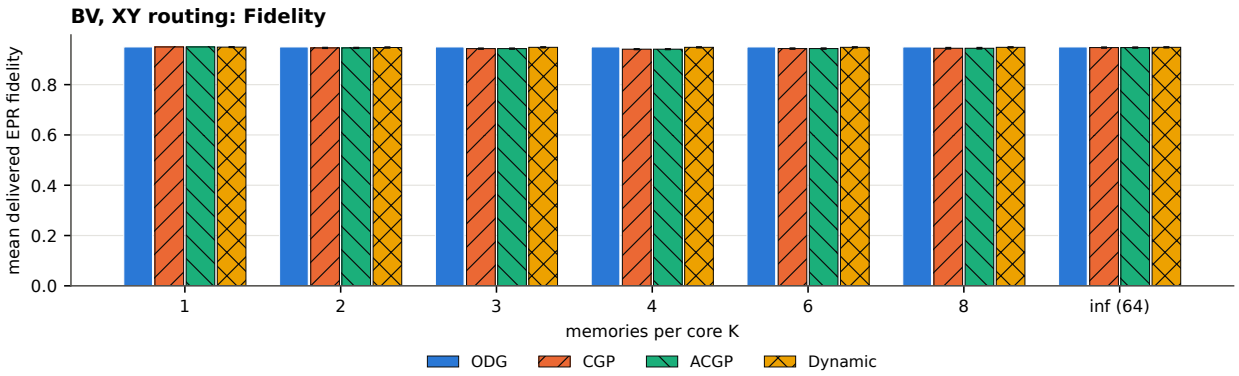



**Figure 2.18:** BV, XY routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 23.9 | 45.7 | 67.4 | 84.8 | 73.9 | 93.5 |
| ACGP | 0.0 | 23.9 | 45.7 | 67.4 | 84.8 | 73.9 | 93.5 |
| Dynamic | 43.5 | 63.0 | 65.2 | 65.2 | 65.2 | 65.2 | 65.2 |

**Table 2.19:** BV, XY routing: coverage (hops with a ready pair (%)) for every K.

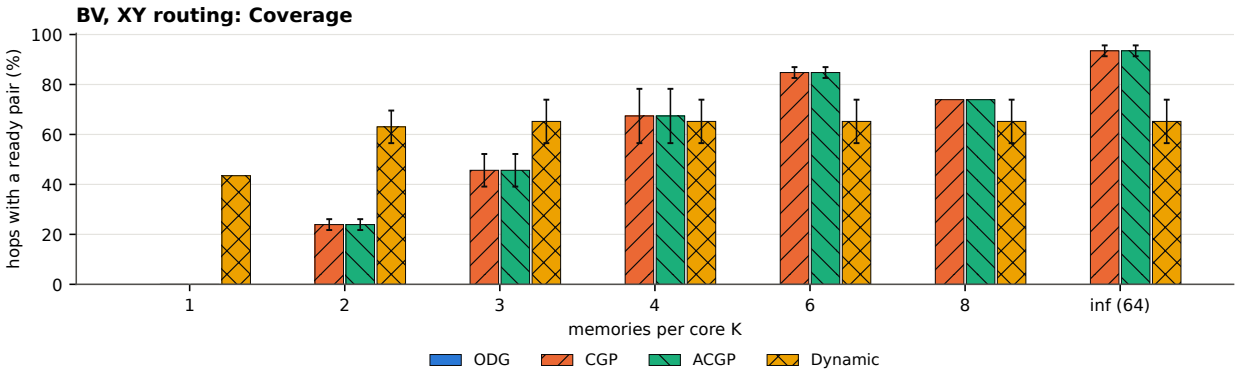



**Figure 2.19:** BV, XY routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 46.43 | 54.19 | 56.67 | 57.98 | 72.36 | 90.97 |
| ACGP | – | 46.43 | 54.19 | 56.67 | 57.98 | 72.36 | 90.97 |
| Dynamic | 37.50 | 15.00 | 5.56 | 5.56 | 5.56 | 5.56 | 5.56 |

**Table 2.20:** BV, XY routing: wastage (pairs generated but not used (%)) for every K.

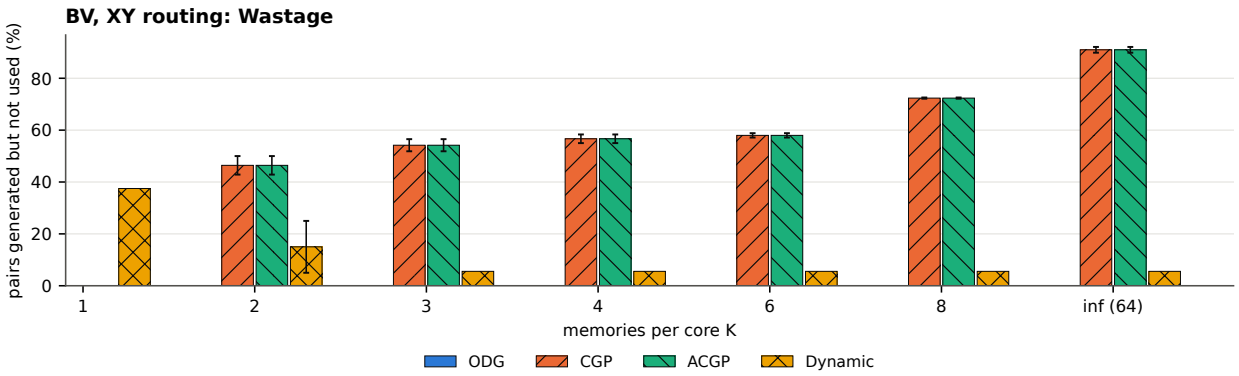



**Figure 2.20:** BV, XY routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 53.6 | 45.8 | 43.3 | 42.0 | 27.6 | 9.0 |
| ACGP | – | 53.6 | 45.8 | 43.3 | 42.0 | 27.6 | 9.0 |
| Dynamic | 62.5 | 85.0 | 94.4 | 94.4 | 94.4 | 94.4 | 94.4 |

**Table 2.21:** BV, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

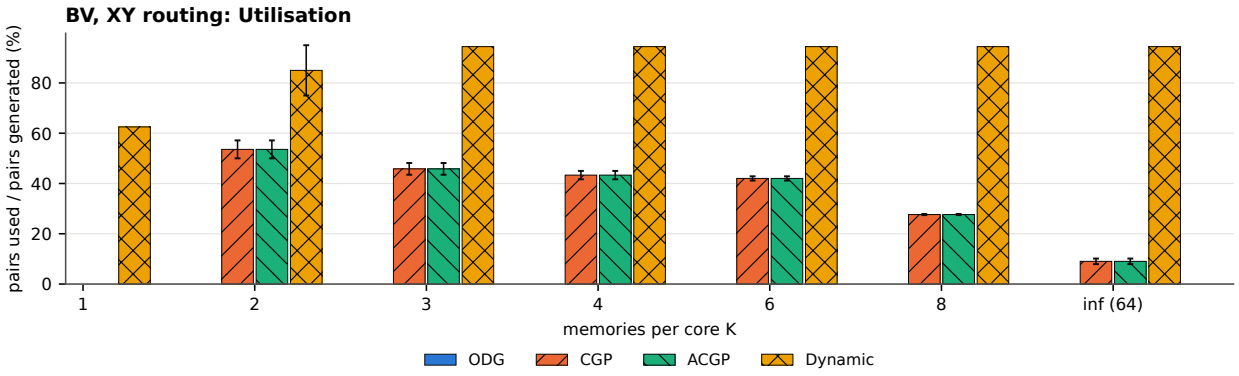



**Figure 2.21:** BV, XY routing: utilisation for every K.

## 2.4 BV, history routing

23 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.4.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.550 | 3.40 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.648 | 1.60 | 0.9447 | 73.9 | 60.7 | 72.36 | 27.6 |
| ACGP | 0.648 | 1.60 | 0.9447 | 73.9 | 60.7 | 72.36 | 27.6 |
| Dynamic (8) | 0.691 | 1.70 | 0.9484 | 65.2 | 53.6 | 5.56 | 94.4 |
| Dynamic ($\infty$) | 0.691 | 1.70 | 0.9484 | 65.2 | 53.6 | 5.56 | 94.4 |

**Table 2.22:** BV, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

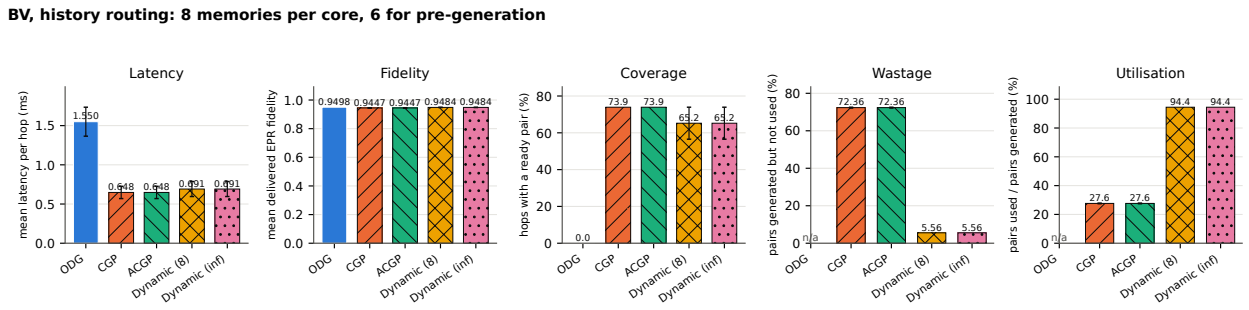



**Figure 2.22:** BV, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.4.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.550 | 3.40 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.811 | 2.40 | 0.9410 | 67.4 | 46.4 | 56.67 | 43.3 |
| ACGP | 0.811 | 2.40 | 0.9410 | 67.4 | 46.4 | 56.67 | 43.3 |
| Dynamic (4) | 0.691 | 1.70 | 0.9484 | 65.2 | 53.6 | 5.56 | 94.4 |
| Dynamic ($\infty$) | 0.691 | 1.70 | 0.9484 | 65.2 | 53.6 | 5.56 | 94.4 |

**Table 2.23:** BV, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

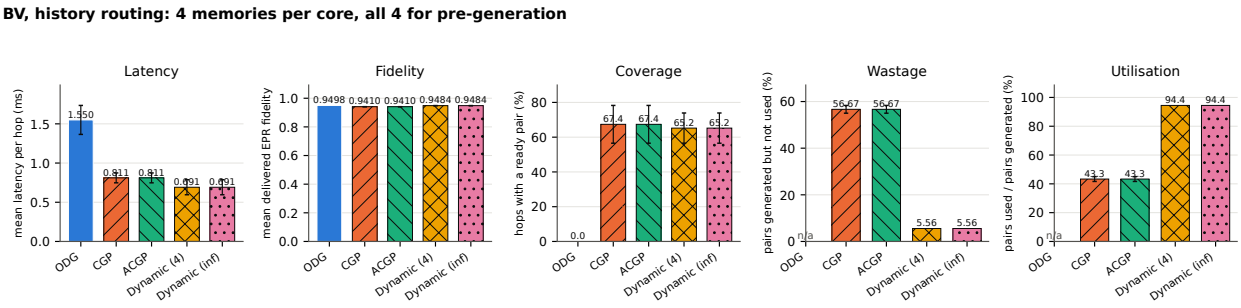



**Figure 2.23:** BV, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.4.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 1.550 | 1.550 | 1.550 | 1.550 | 1.550 | 1.550 | 1.550 |
| CGP | 1.550 | 1.228 | 0.948 | 0.811 | 0.526 | 0.648 | 0.504 |
| ACGP | 1.550 | 1.228 | 0.948 | 0.811 | 0.526 | 0.648 | 0.504 |
| Dynamic | 2.065 | 1.065 | 0.691 | 0.691 | 0.691 | 0.691 | 0.691 |

**Table 2.24:** BV, history routing: latency (mean latency per hop (ms)) for every K.

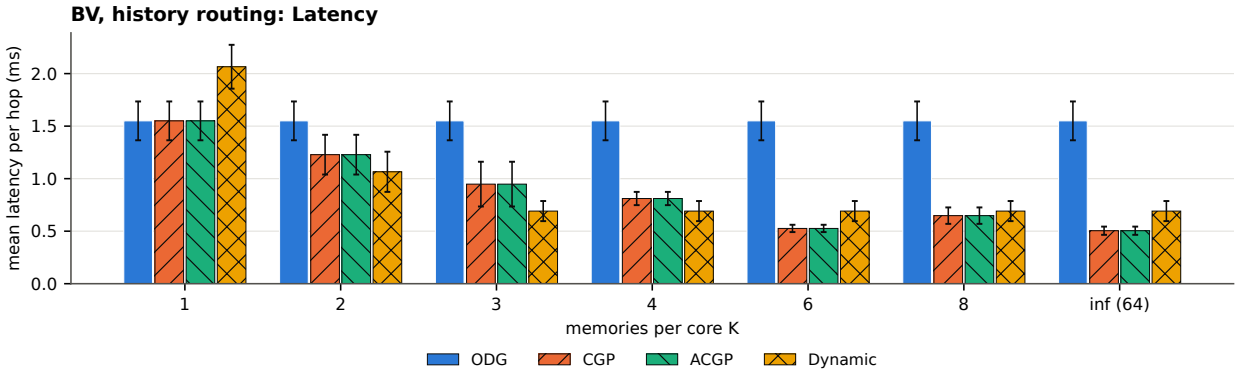



**Figure 2.24:** BV, history routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9460 | 0.9432 | 0.9410 | 0.9434 | 0.9447 | 0.9473 |
| ACGP | 0.9498 | 0.9460 | 0.9432 | 0.9410 | 0.9434 | 0.9447 | 0.9473 |
| Dynamic | 0.9493 | 0.9477 | 0.9484 | 0.9484 | 0.9484 | 0.9484 | 0.9484 |

**Table 2.25:** BV, history routing: fidelity (mean delivered EPR fidelity) for every K.

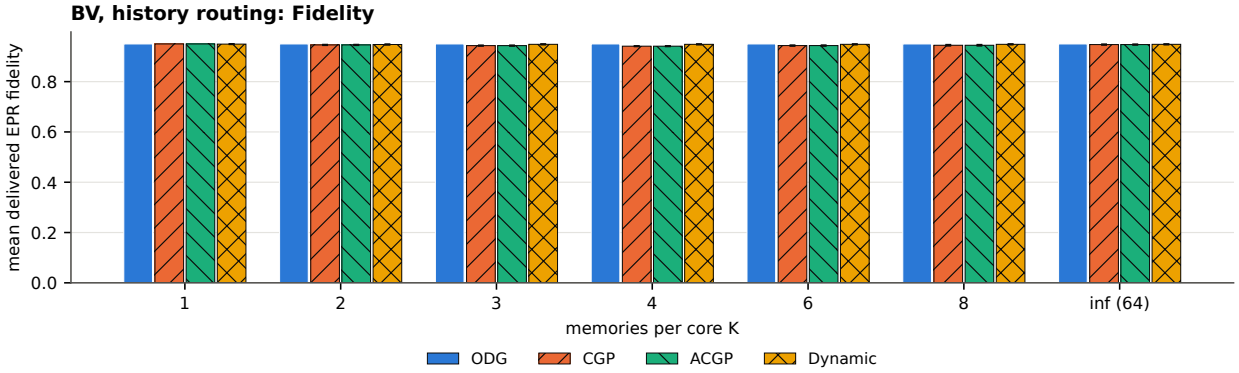



**Figure 2.25:** BV, history routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 23.9 | 45.7 | 67.4 | 84.8 | 73.9 | 93.5 |
| ACGP | 0.0 | 23.9 | 45.7 | 67.4 | 84.8 | 73.9 | 93.5 |
| Dynamic | 43.5 | 63.0 | 65.2 | 65.2 | 65.2 | 65.2 | 65.2 |

**Table 2.26:** BV, history routing: coverage (hops with a ready pair (%)) for every K.

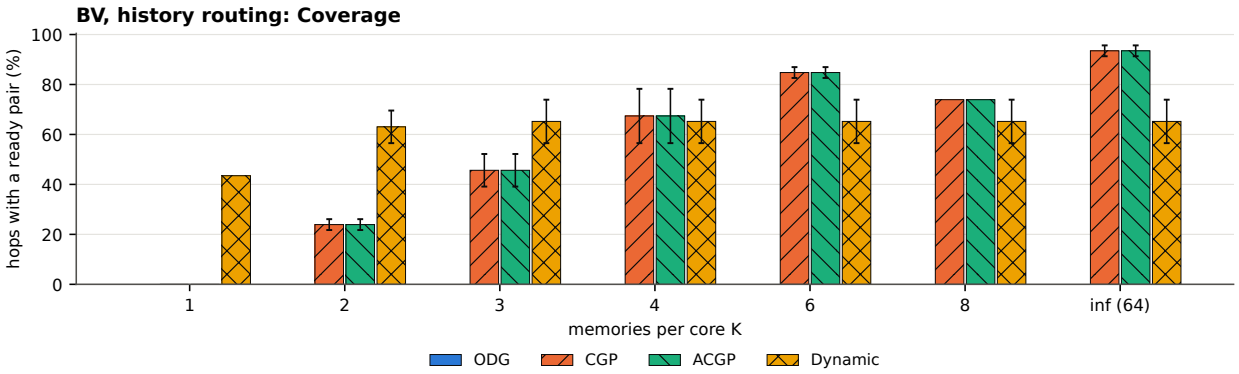



**Figure 2.26:** BV, history routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 46.43 | 54.19 | 56.67 | 57.98 | 72.36 | 90.97 |
| ACGP | – | 46.43 | 54.19 | 56.67 | 57.98 | 72.36 | 90.97 |
| Dynamic | 37.50 | 15.00 | 5.56 | 5.56 | 5.56 | 5.56 | 5.56 |

**Table 2.27:** BV, history routing: wastage (pairs generated but not used (%)) for every K.

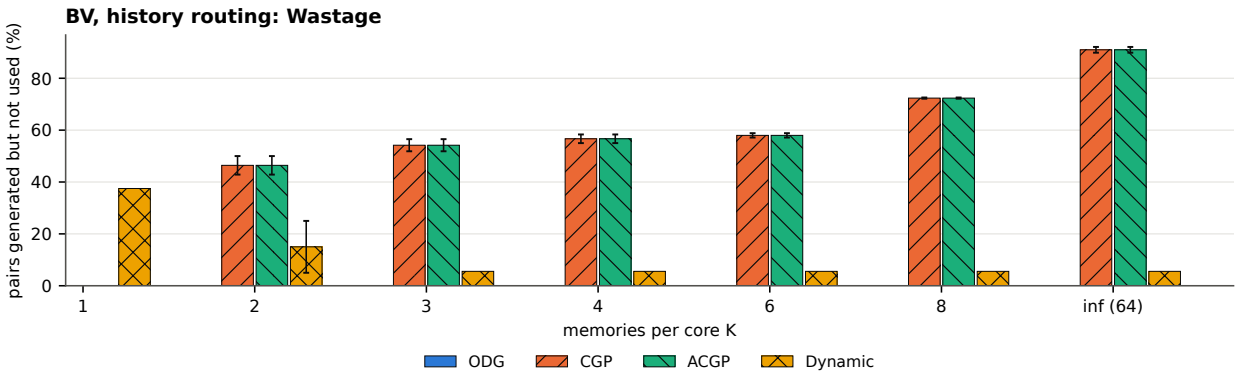



**Figure 2.27:** BV, history routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 53.6 | 45.8 | 43.3 | 42.0 | 27.6 | 9.0 |
| ACGP | – | 53.6 | 45.8 | 43.3 | 42.0 | 27.6 | 9.0 |
| Dynamic | 62.5 | 85.0 | 94.4 | 94.4 | 94.4 | 94.4 | 94.4 |

**Table 2.28:** BV, history routing: utilisation (pairs used / pairs generated (%)) for every K.

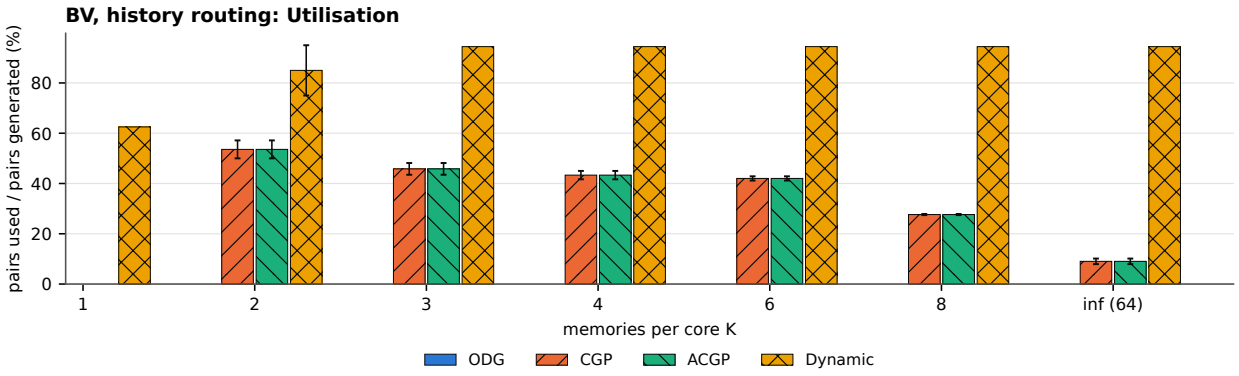



**Figure 2.28:** BV, history routing: utilisation for every K.

## 2.5 GHZ, XY routing

24 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.5.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.923 | 4.20 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.633 | 1.45 | 0.9432 | 79.2 | 70.0 | 69.81 | 30.2 |
| ACGP | 0.633 | 1.45 | 0.9432 | 79.2 | 70.0 | 69.81 | 30.2 |
| Dynamic (8) | 0.865 | 2.00 | 0.9481 | 60.4 | 40.0 | 2.50 | 97.5 |
| Dynamic ($\infty$) | 0.865 | 2.00 | 0.9481 | 60.4 | 40.0 | 2.50 | 97.5 |

**Table 2.29:** GHZ, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

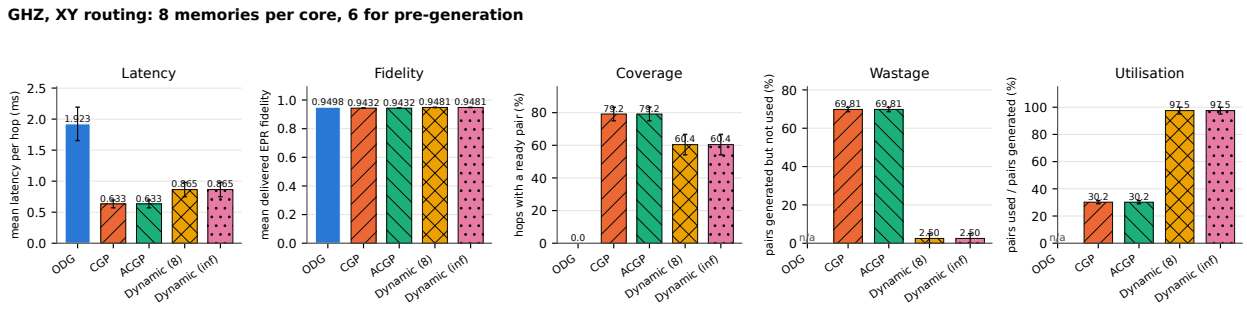



**Figure 2.29:** GHZ, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.5.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.923 | 4.20 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.940 | 3.15 | 0.9412 | 64.6 | 46.7 | 55.71 | 44.3 |
| ACGP | 0.940 | 3.15 | 0.9412 | 64.6 | 46.7 | 55.71 | 44.3 |
| Dynamic (4) | 0.865 | 2.00 | 0.9481 | 60.4 | 40.0 | 2.50 | 97.5 |
| Dynamic ($\infty$) | 0.865 | 2.00 | 0.9481 | 60.4 | 40.0 | 2.50 | 97.5 |

**Table 2.30:** GHZ, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

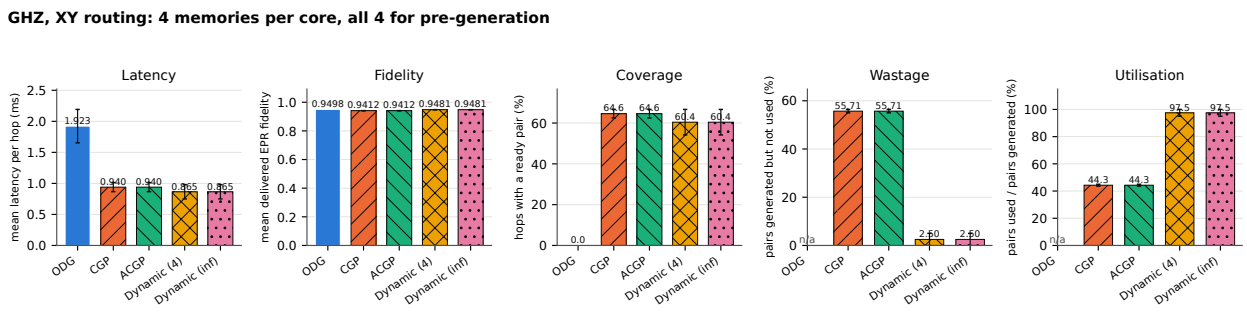



**Figure 2.30:** GHZ, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.5.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 1.923 | 1.923 | 1.923 | 1.923 | 1.923 | 1.923 | 1.923 |
| CGP | 1.923 | 1.623 | 1.225 | 0.940 | 0.671 | 0.633 | 0.558 |
| ACGP | 1.923 | 1.623 | 1.225 | 0.940 | 0.671 | 0.633 | 0.558 |
| Dynamic | 2.406 | 1.260 | 0.865 | 0.865 | 0.865 | 0.865 | 0.865 |

**Table 2.31:** GHZ, XY routing: latency (mean latency per hop (ms)) for every K.

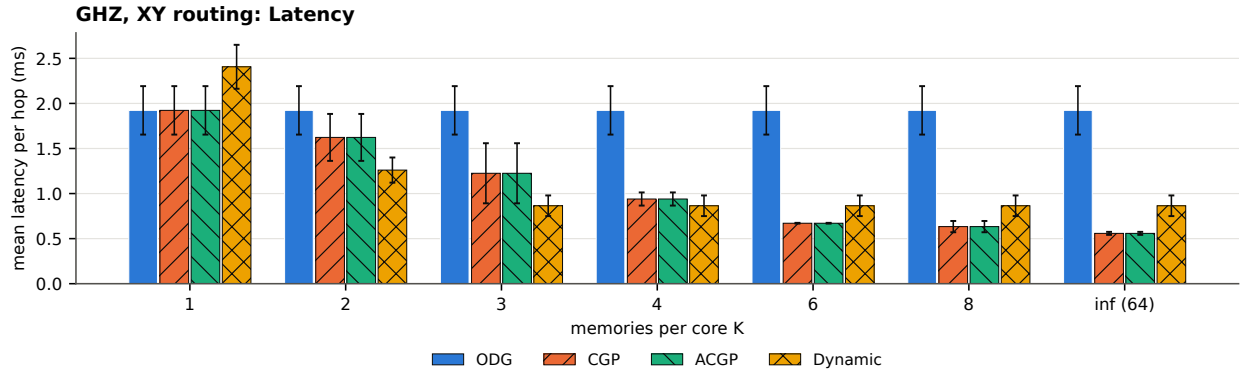



**Figure 2.31:** GHZ, XY routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9442 | 0.9436 | 0.9412 | 0.9437 | 0.9432 | 0.9471 |
| ACGP | 0.9498 | 0.9442 | 0.9436 | 0.9412 | 0.9437 | 0.9432 | 0.9471 |
| Dynamic | 0.9489 | 0.9479 | 0.9481 | 0.9481 | 0.9481 | 0.9481 | 0.9481 |

**Table 2.32:** GHZ, XY routing: fidelity (mean delivered EPR fidelity) for every K.

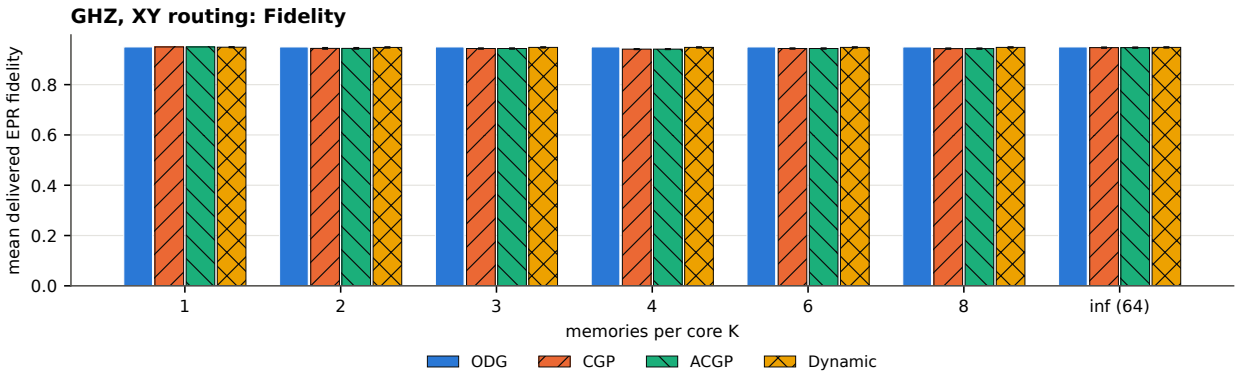



**Figure 2.32:** GHZ, XY routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 25.0 | 47.9 | 64.6 | 79.2 | 79.2 | 89.6 |
| ACGP | 0.0 | 25.0 | 47.9 | 64.6 | 79.2 | 79.2 | 89.6 |
| Dynamic | 45.8 | 56.2 | 60.4 | 60.4 | 60.4 | 60.4 | 60.4 |

**Table 2.33:** GHZ, XY routing: coverage (hops with a ready pair (%)) for every K.

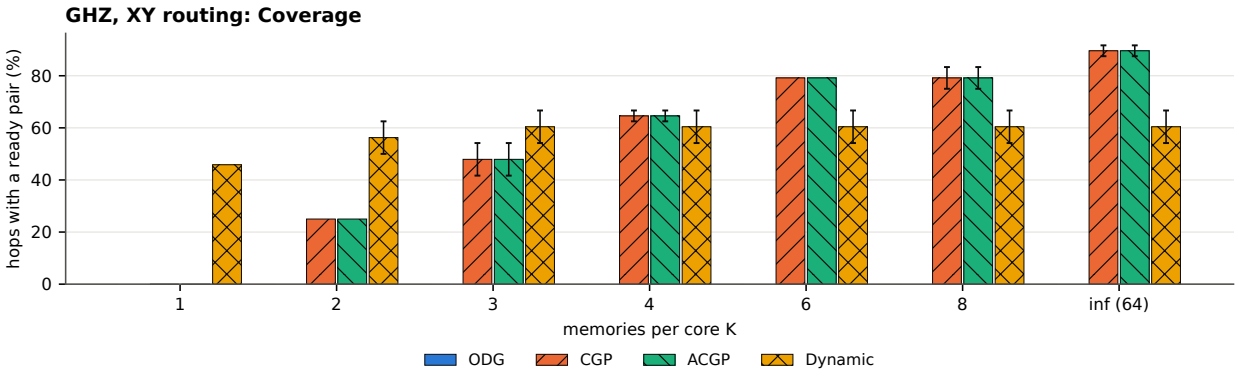



**Figure 2.33:** GHZ, XY routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 46.41 | 53.00 | 55.71 | 58.82 | 69.81 | 91.75 |
| ACGP | – | 46.41 | 53.00 | 55.71 | 58.82 | 69.81 | 91.75 |
| Dynamic | 35.29 | 22.04 | 2.50 | 2.50 | 2.50 | 2.50 | 2.50 |

**Table 2.34:** GHZ, XY routing: wastage (pairs generated but not used (%)) for every K.

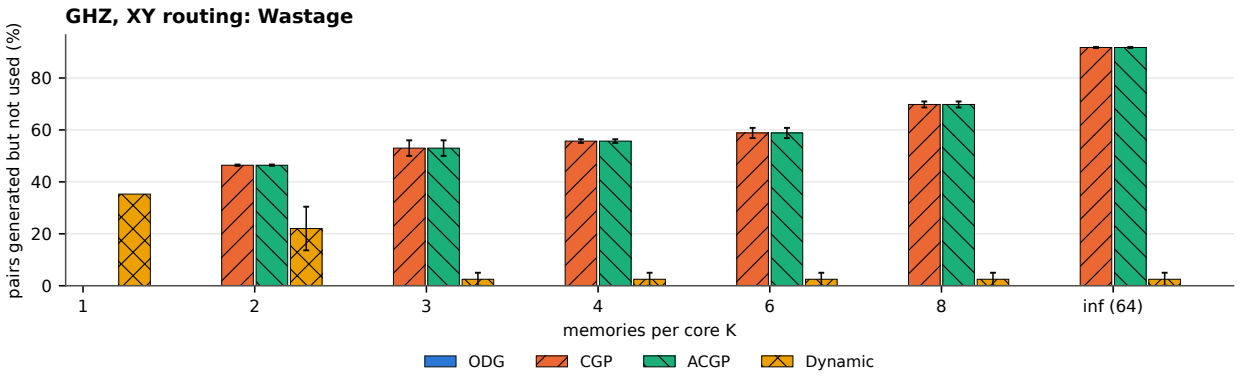



**Figure 2.34:** GHZ, XY routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 53.6 | 47.0 | 44.3 | 41.2 | 30.2 | 8.3 |
| ACGP | – | 53.6 | 47.0 | 44.3 | 41.2 | 30.2 | 8.3 |
| Dynamic | 64.7 | 78.0 | 97.5 | 97.5 | 97.5 | 97.5 | 97.5 |

**Table 2.35:** GHZ, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

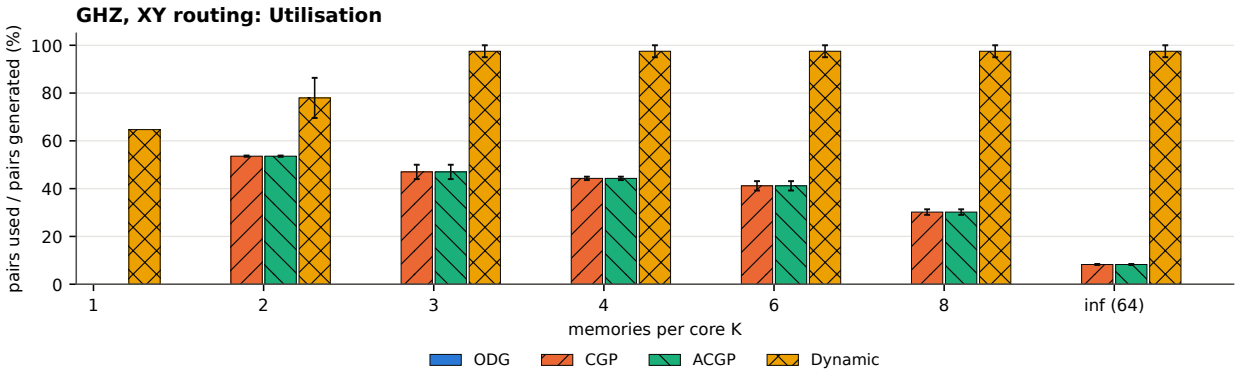



**Figure 2.35:** GHZ, XY routing: utilisation for every K.

## 2.6 GHZ, history routing

24 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.6.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.923 | 4.20 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.633 | 1.45 | 0.9432 | 79.2 | 70.0 | 69.81 | 30.2 |
| ACGP | 0.633 | 1.45 | 0.9432 | 79.2 | 70.0 | 69.81 | 30.2 |
| Dynamic (8) | 0.865 | 2.00 | 0.9481 | 60.4 | 40.0 | 2.50 | 97.5 |
| Dynamic ($\infty$) | 0.865 | 2.00 | 0.9481 | 60.4 | 40.0 | 2.50 | 97.5 |

**Table 2.36:** GHZ, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

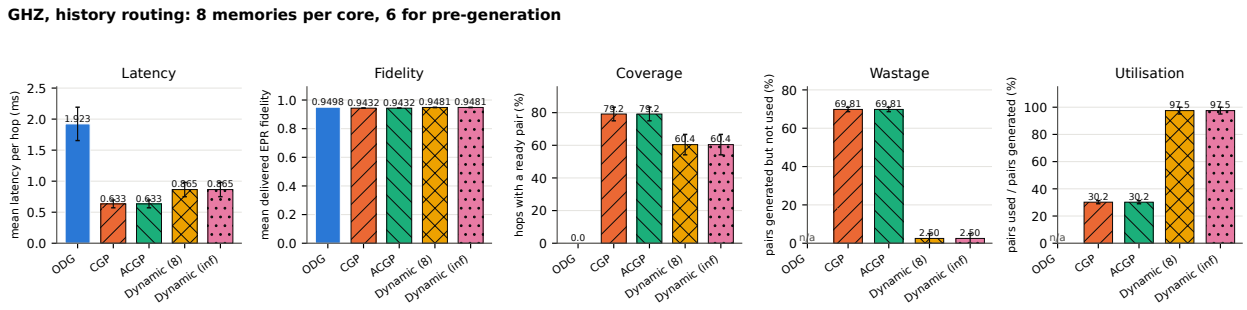



**Figure 2.36:** GHZ, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.6.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.923 | 4.20 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.940 | 3.15 | 0.9412 | 64.6 | 46.7 | 55.71 | 44.3 |
| ACGP | 0.940 | 3.15 | 0.9412 | 64.6 | 46.7 | 55.71 | 44.3 |
| Dynamic (4) | 0.865 | 2.00 | 0.9481 | 60.4 | 40.0 | 2.50 | 97.5 |
| Dynamic ($\infty$) | 0.865 | 2.00 | 0.9481 | 60.4 | 40.0 | 2.50 | 97.5 |

**Table 2.37:** GHZ, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

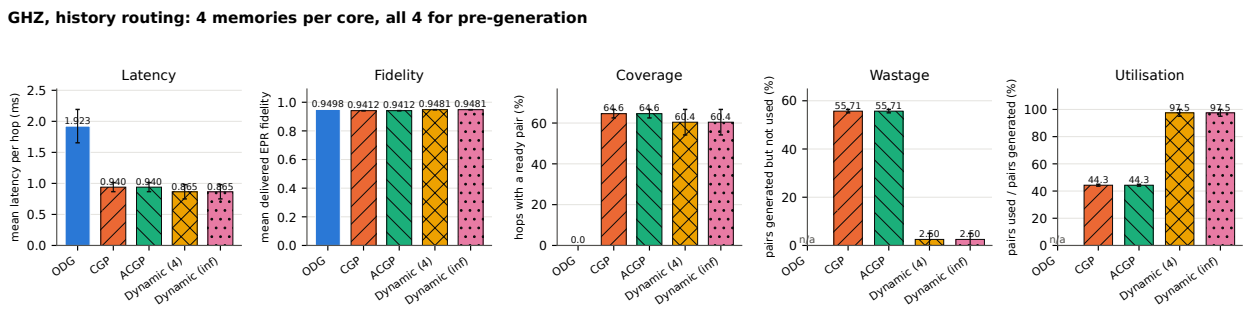



**Figure 2.37:** GHZ, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.6.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 1.923 | 1.923 | 1.923 | 1.923 | 1.923 | 1.923 | 1.923 |
| CGP | 1.923 | 1.623 | 1.225 | 0.940 | 0.671 | 0.633 | 0.558 |
| ACGP | 1.923 | 1.623 | 1.225 | 0.940 | 0.671 | 0.633 | 0.558 |
| Dynamic | 2.406 | 1.260 | 0.865 | 0.865 | 0.865 | 0.865 | 0.865 |

**Table 2.38:** GHZ, history routing: latency (mean latency per hop (ms)) for every K.

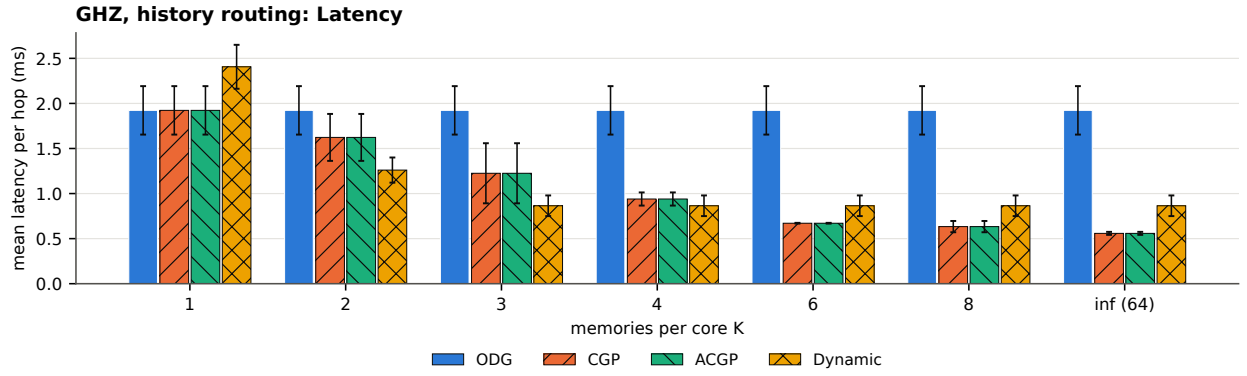



**Figure 2.38:** GHZ, history routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9442 | 0.9436 | 0.9412 | 0.9437 | 0.9432 | 0.9471 |
| ACGP | 0.9498 | 0.9442 | 0.9436 | 0.9412 | 0.9437 | 0.9432 | 0.9471 |
| Dynamic | 0.9489 | 0.9479 | 0.9481 | 0.9481 | 0.9481 | 0.9481 | 0.9481 |

**Table 2.39:** GHZ, history routing: fidelity (mean delivered EPR fidelity) for every K.

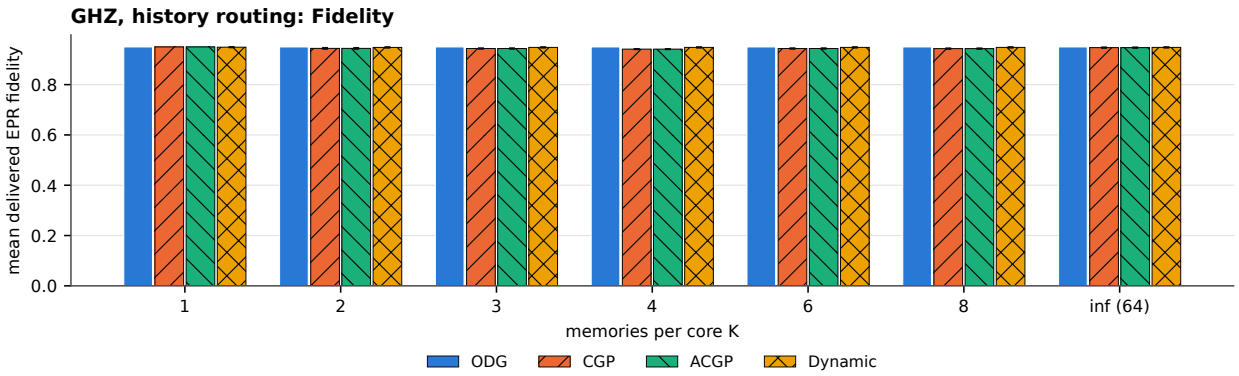



**Figure 2.39:** GHZ, history routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 25.0 | 47.9 | 64.6 | 79.2 | 79.2 | 89.6 |
| ACGP | 0.0 | 25.0 | 47.9 | 64.6 | 79.2 | 79.2 | 89.6 |
| Dynamic | 45.8 | 56.2 | 60.4 | 60.4 | 60.4 | 60.4 | 60.4 |

**Table 2.40:** GHZ, history routing: coverage (hops with a ready pair (%)) for every K.

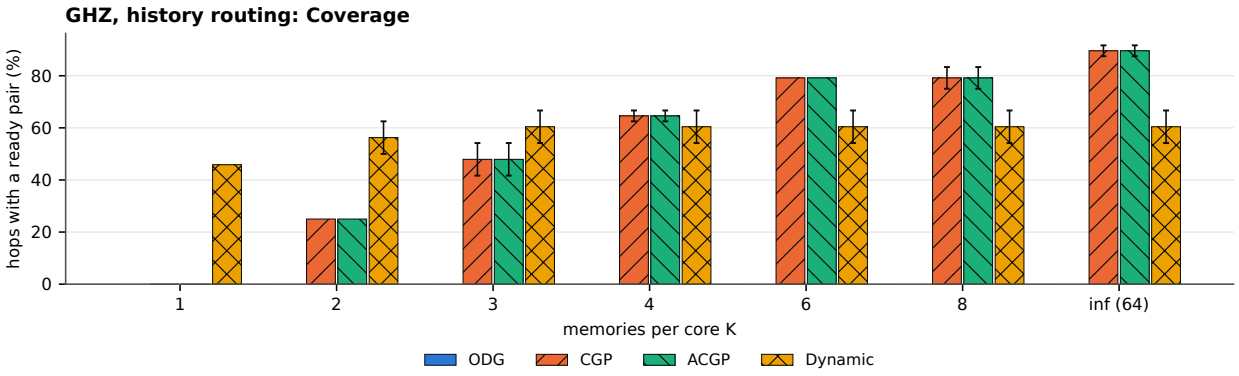



**Figure 2.40:** GHZ, history routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 46.41 | 53.00 | 55.71 | 58.82 | 69.81 | 91.75 |
| ACGP | – | 46.41 | 53.00 | 55.71 | 58.82 | 69.81 | 91.75 |
| Dynamic | 35.29 | 22.04 | 2.50 | 2.50 | 2.50 | 2.50 | 2.50 |

**Table 2.41:** GHZ, history routing: wastage (pairs generated but not used (%)) for every K.

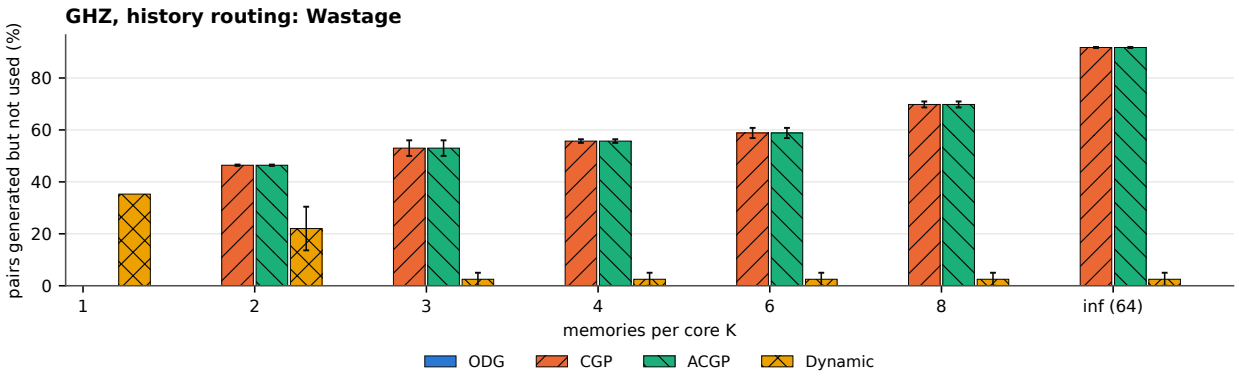



**Figure 2.41:** GHZ, history routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 53.6 | 47.0 | 44.3 | 41.2 | 30.2 | 8.3 |
| ACGP | – | 53.6 | 47.0 | 44.3 | 41.2 | 30.2 | 8.3 |
| Dynamic | 64.7 | 78.0 | 97.5 | 97.5 | 97.5 | 97.5 | 97.5 |

**Table 2.42:** GHZ, history routing: utilisation (pairs used / pairs generated (%)) for every K.

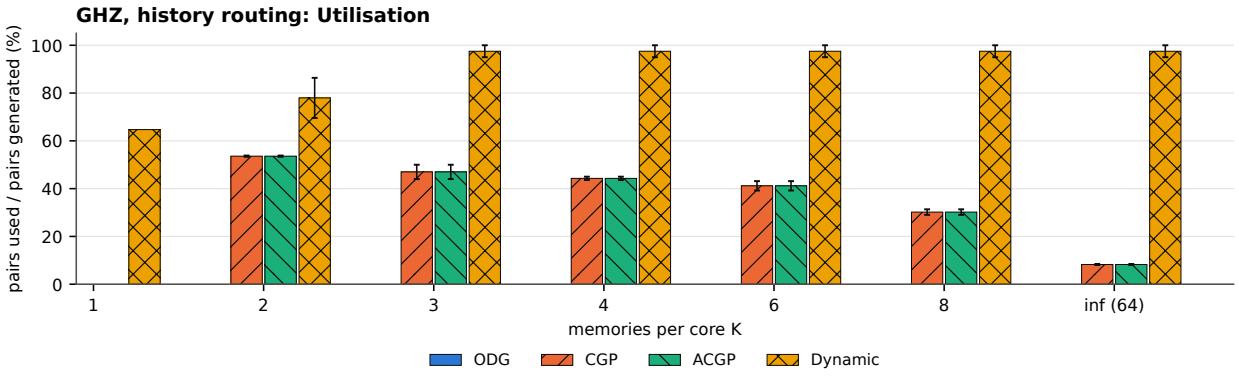



**Figure 2.42:** GHZ, history routing: utilisation for every K.

## 2.7 DJ, XY routing

30 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.7.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.777 | 3.95 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.643 | 1.45 | 0.9427 | 71.7 | 62.5 | 65.73 | 34.3 |
| ACGP | 0.643 | 1.45 | 0.9427 | 71.7 | 62.5 | 65.73 | 34.3 |
| Dynamic (8) | 1.008 | 2.85 | 0.9478 | 56.7 | 53.1 | 2.50 | 97.5 |
| Dynamic ($\infty$) | 1.008 | 2.85 | 0.9478 | 56.7 | 53.1 | 2.50 | 97.5 |

**Table 2.43:** DJ, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

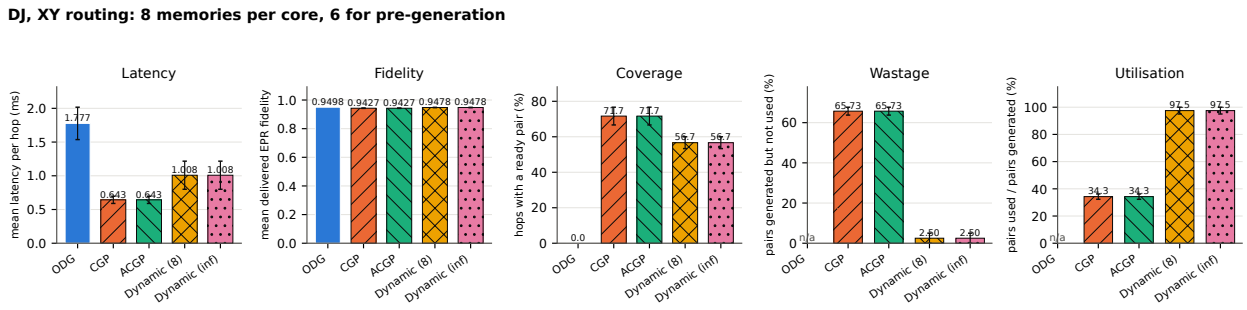



**Figure 2.43:** DJ, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.7.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.777 | 3.95 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.858 | 3.10 | 0.9377 | 68.3 | 50.0 | 49.44 | 50.6 |
| ACGP | 0.858 | 3.10 | 0.9377 | 68.3 | 50.0 | 49.44 | 50.6 |
| Dynamic (4) | 1.008 | 2.85 | 0.9478 | 56.7 | 53.1 | 2.50 | 97.5 |
| Dynamic ($\infty$) | 1.008 | 2.85 | 0.9478 | 56.7 | 53.1 | 2.50 | 97.5 |

**Table 2.44:** DJ, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

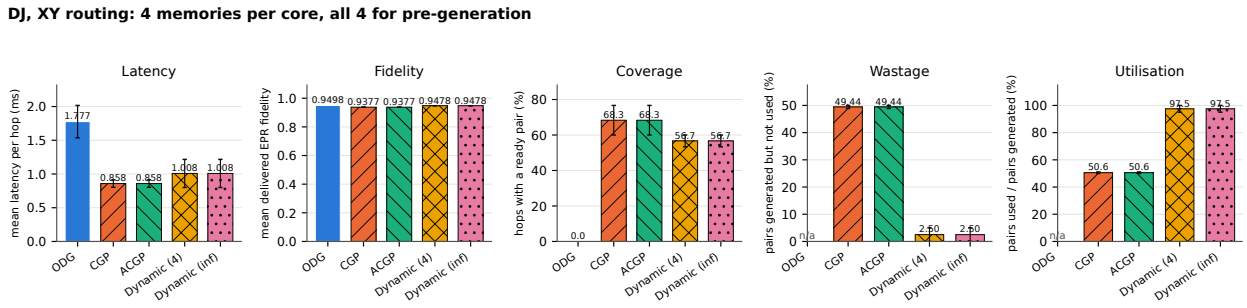



**Figure 2.44:** DJ, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.7.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 1.777 | 1.777 | 1.777 | 1.777 | 1.777 | 1.777 | 1.777 |
| CGP | 1.777 | 1.457 | 1.152 | 0.858 | 0.612 | 0.643 | 0.508 |
| ACGP | 1.777 | 1.457 | 1.152 | 0.858 | 0.612 | 0.643 | 0.508 |
| Dynamic | 2.167 | 1.375 | 1.008 | 1.008 | 1.008 | 1.008 | 1.008 |

**Table 2.45:** DJ, XY routing: latency (mean latency per hop (ms)) for every K.

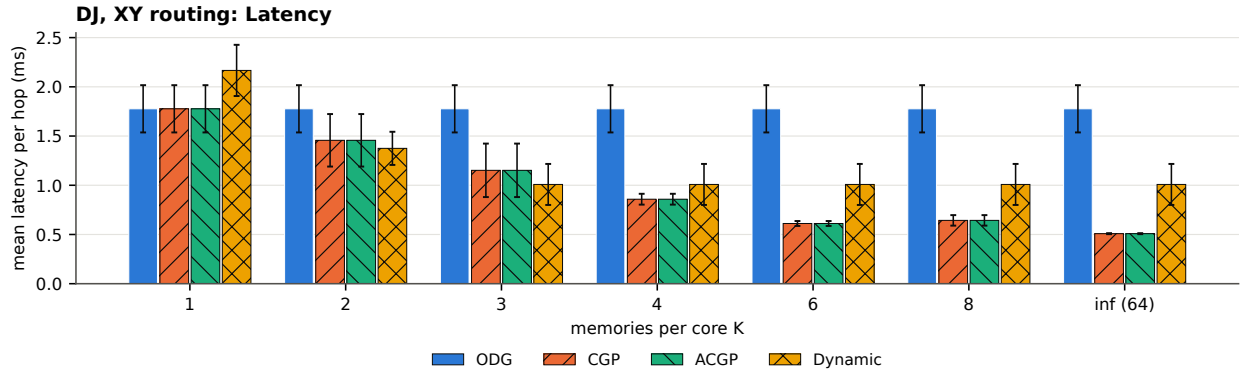



**Figure 2.45:** DJ, XY routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9433 | 0.9386 | 0.9377 | 0.9392 | 0.9427 | 0.9473 |
| ACGP | 0.9498 | 0.9433 | 0.9386 | 0.9377 | 0.9392 | 0.9427 | 0.9473 |
| Dynamic | 0.9493 | 0.9470 | 0.9478 | 0.9478 | 0.9478 | 0.9478 | 0.9478 |

**Table 2.46:** DJ, XY routing: fidelity (mean delivered EPR fidelity) for every K.

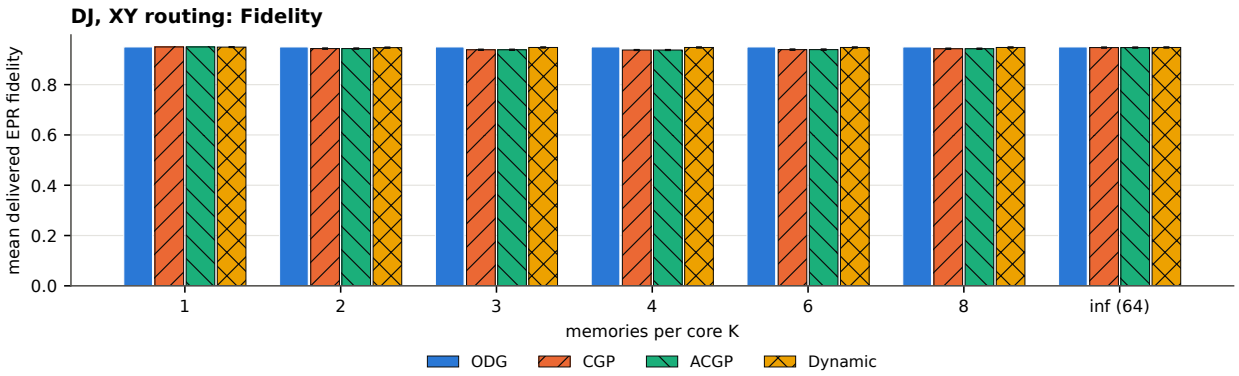



**Figure 2.46:** DJ, XY routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 26.7 | 45.0 | 68.3 | 80.0 | 71.7 | 90.0 |
| ACGP | 0.0 | 26.7 | 45.0 | 68.3 | 80.0 | 71.7 | 90.0 |
| Dynamic | 36.7 | 46.7 | 56.7 | 56.7 | 56.7 | 56.7 | 56.7 |

**Table 2.47:** DJ, XY routing: coverage (hops with a ready pair (%)) for every K.

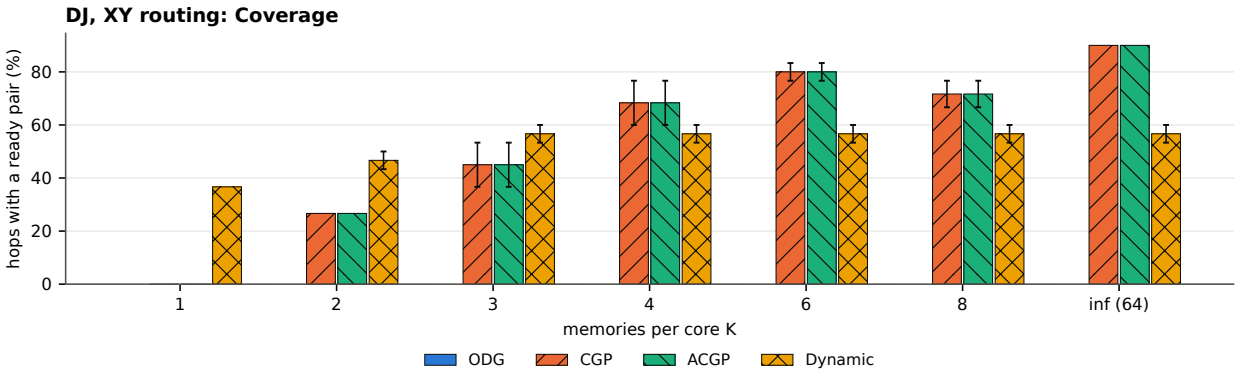



**Figure 2.47:** DJ, XY routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 40.18 | 47.88 | 49.44 | 52.33 | 65.73 | 91.10 |
| ACGP | – | 40.18 | 47.88 | 49.44 | 52.33 | 65.73 | 91.10 |
| Dynamic | 38.89 | 14.64 | 2.50 | 2.50 | 2.50 | 2.50 | 2.50 |

**Table 2.48:** DJ, XY routing: wastage (pairs generated but not used (%)) for every K.

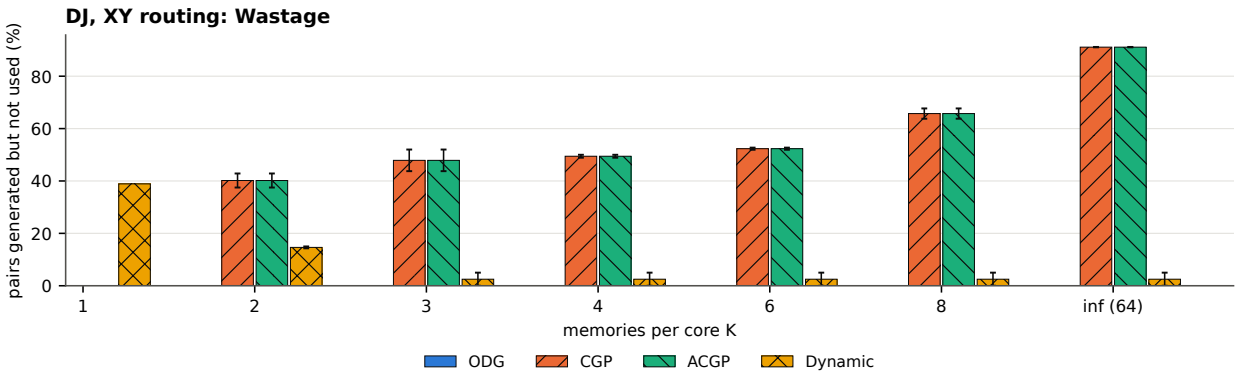



**Figure 2.48:** DJ, XY routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 59.8 | 52.1 | 50.6 | 47.7 | 34.3 | 8.9 |
| ACGP | – | 59.8 | 52.1 | 50.6 | 47.7 | 34.3 | 8.9 |
| Dynamic | 61.1 | 85.4 | 97.5 | 97.5 | 97.5 | 97.5 | 97.5 |

**Table 2.49:** DJ, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

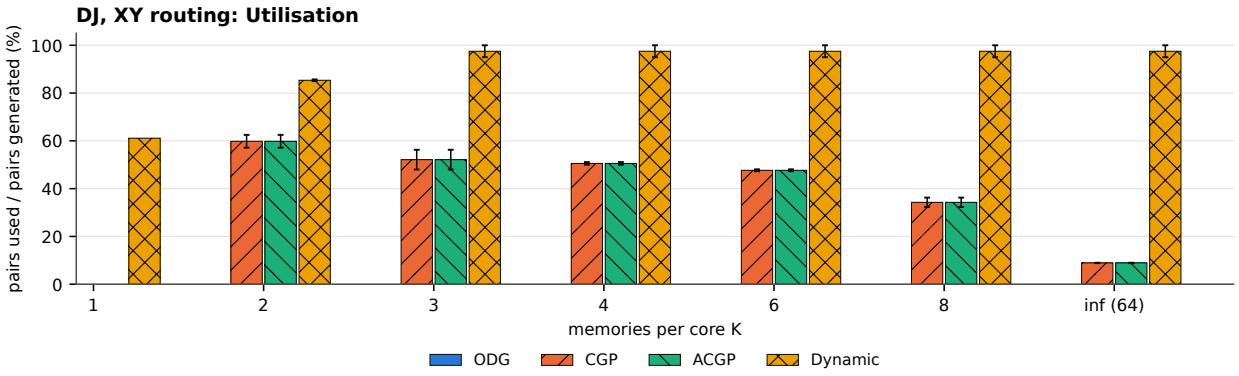



**Figure 2.49:** DJ, XY routing: utilisation for every K.

## 2.8 DJ, history routing

30 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.8.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.777 | 3.95 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.643 | 1.45 | 0.9427 | 71.7 | 62.5 | 65.73 | 34.3 |
| ACGP | 0.643 | 1.45 | 0.9427 | 71.7 | 62.5 | 65.73 | 34.3 |
| Dynamic (8) | 1.008 | 2.85 | 0.9478 | 56.7 | 53.1 | 2.50 | 97.5 |
| Dynamic ($\infty$) | 1.008 | 2.85 | 0.9478 | 56.7 | 53.1 | 2.50 | 97.5 |

**Table 2.50:** DJ, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

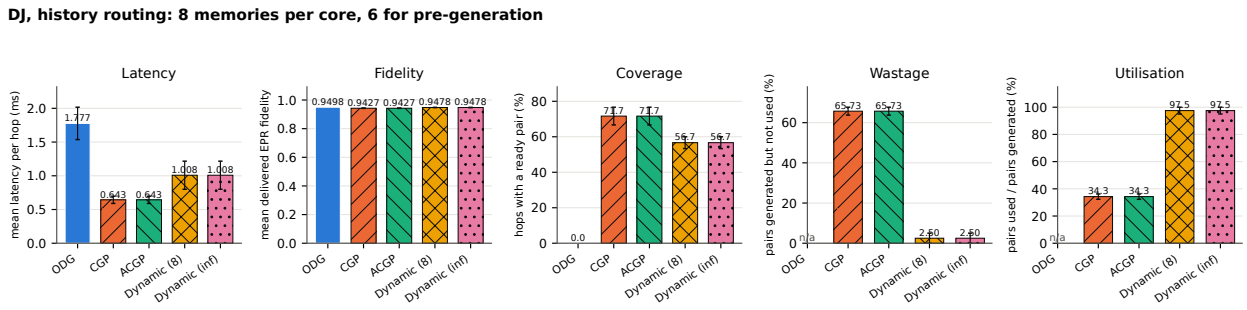



**Figure 2.50:** DJ, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.8.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.777 | 3.95 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.858 | 3.10 | 0.9377 | 68.3 | 50.0 | 49.44 | 50.6 |
| ACGP | 0.858 | 3.10 | 0.9377 | 68.3 | 50.0 | 49.44 | 50.6 |
| Dynamic (4) | 1.008 | 2.85 | 0.9478 | 56.7 | 53.1 | 2.50 | 97.5 |
| Dynamic ($\infty$) | 1.008 | 2.85 | 0.9478 | 56.7 | 53.1 | 2.50 | 97.5 |

**Table 2.51:** DJ, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

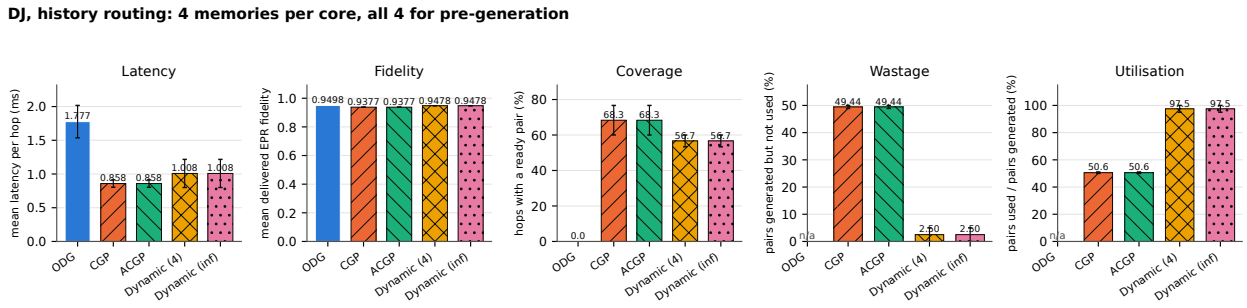



**Figure 2.51:** DJ, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.8.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 1.777 | 1.777 | 1.777 | 1.777 | 1.777 | 1.777 | 1.777 |
| CGP | 1.777 | 1.457 | 1.152 | 0.858 | 0.612 | 0.643 | 0.508 |
| ACGP | 1.777 | 1.457 | 1.152 | 0.858 | 0.612 | 0.643 | 0.508 |
| Dynamic | 2.167 | 1.375 | 1.008 | 1.008 | 1.008 | 1.008 | 1.008 |

**Table 2.52:** DJ, history routing: latency (mean latency per hop (ms)) for every K.

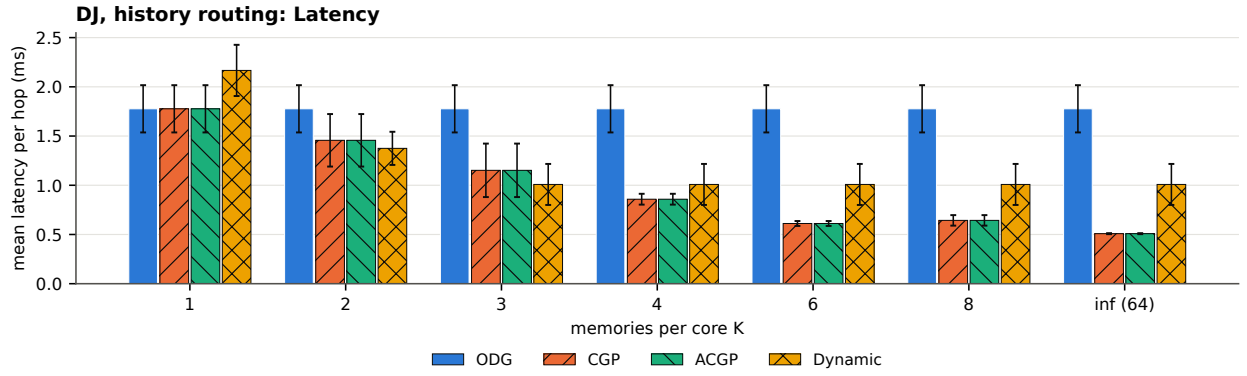



**Figure 2.52:** DJ, history routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9433 | 0.9386 | 0.9377 | 0.9392 | 0.9427 | 0.9473 |
| ACGP | 0.9498 | 0.9433 | 0.9386 | 0.9377 | 0.9392 | 0.9427 | 0.9473 |
| Dynamic | 0.9493 | 0.9470 | 0.9478 | 0.9478 | 0.9478 | 0.9478 | 0.9478 |

**Table 2.53:** DJ, history routing: fidelity (mean delivered EPR fidelity) for every K.

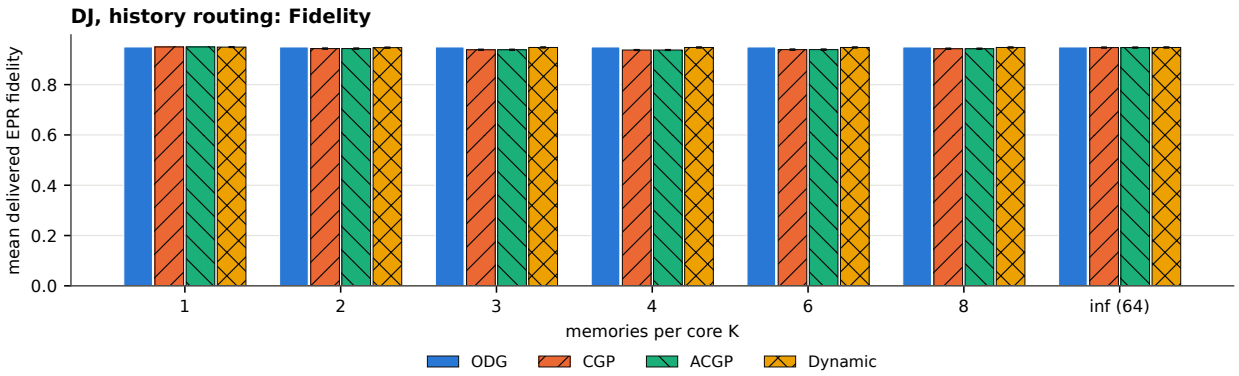



**Figure 2.53:** DJ, history routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 26.7 | 45.0 | 68.3 | 80.0 | 71.7 | 90.0 |
| ACGP | 0.0 | 26.7 | 45.0 | 68.3 | 80.0 | 71.7 | 90.0 |
| Dynamic | 36.7 | 46.7 | 56.7 | 56.7 | 56.7 | 56.7 | 56.7 |

**Table 2.54:** DJ, history routing: coverage (hops with a ready pair (%)) for every K.

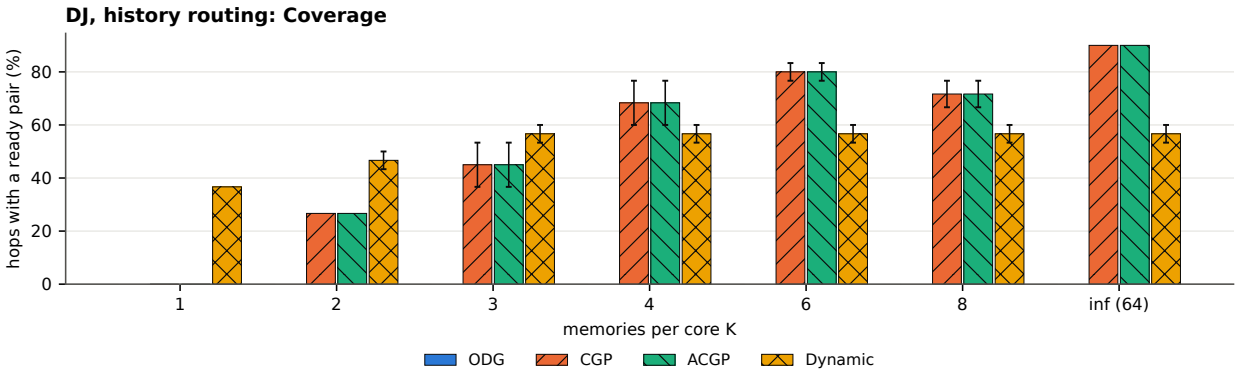



**Figure 2.54:** DJ, history routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 40.18 | 47.88 | 49.44 | 52.33 | 65.73 | 91.10 |
| ACGP | – | 40.18 | 47.88 | 49.44 | 52.33 | 65.73 | 91.10 |
| Dynamic | 38.89 | 14.64 | 2.50 | 2.50 | 2.50 | 2.50 | 2.50 |

**Table 2.55:** DJ, history routing: wastage (pairs generated but not used (%)) for every K.

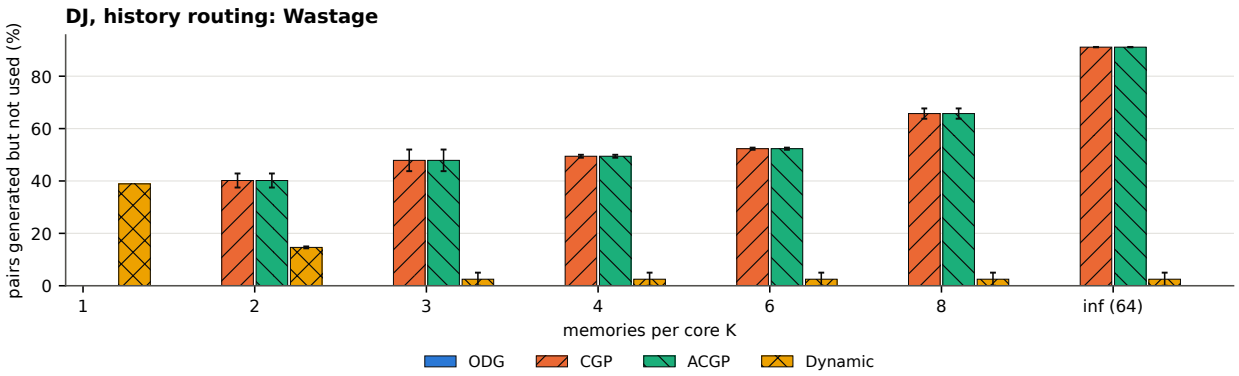



**Figure 2.55:** DJ, history routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 59.8 | 52.1 | 50.6 | 47.7 | 34.3 | 8.9 |
| ACGP | – | 59.8 | 52.1 | 50.6 | 47.7 | 34.3 | 8.9 |
| Dynamic | 61.1 | 85.4 | 97.5 | 97.5 | 97.5 | 97.5 | 97.5 |

**Table 2.56:** DJ, history routing: utilisation (pairs used / pairs generated (%)) for every K.

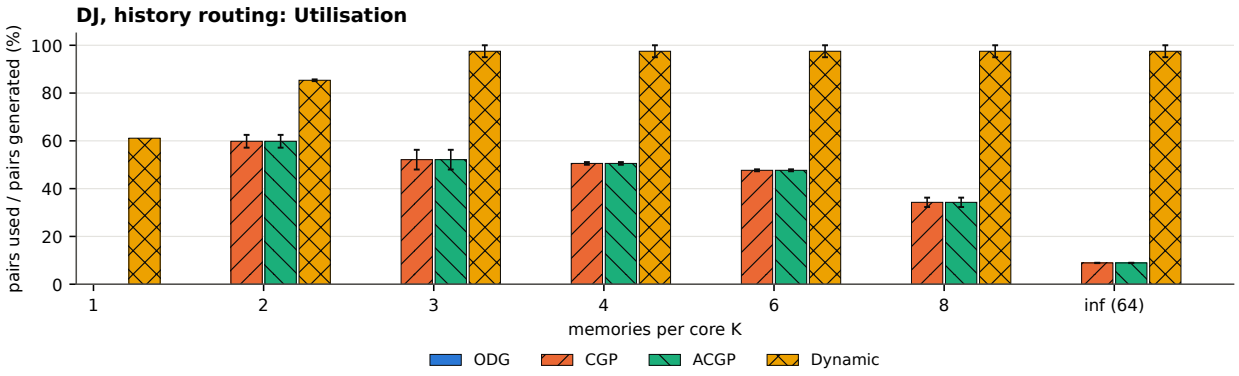



**Figure 2.56:** DJ, history routing: utilisation for every K.

## 2.9 QPE (exact), XY routing

11921 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.9.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.784 | 4.60 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 1.020 | 3.40 | 0.9420 | 75.4 | 54.8 | 4.42 | 95.6 |
| ACGP | 1.016 | 3.35 | 0.9422 | 76.1 | 56.3 | 4.32 | 95.7 |
| Dynamic (8) | 0.758 | 2.40 | 0.9435 | 77.0 | 64.6 | 0.99 | 99.0 |
| Dynamic ($\infty$) | 0.615 | 1.65 | 0.9440 | 84.9 | 72.9 | 1.07 | 98.9 |

**Table 2.57:** QPE (exact), XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

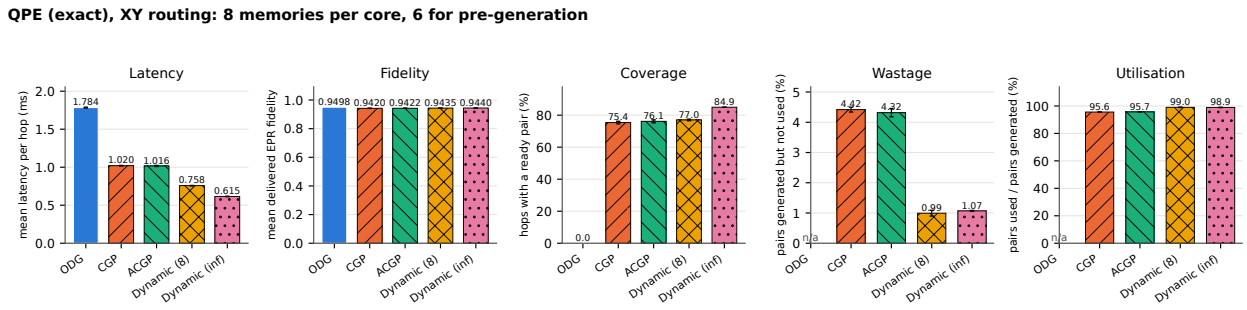



**Figure 2.57:** QPE (exact), XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.9.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.894 | 4.70 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 2.527 | 9.72 | 0.9400 | 61.5 | 36.4 | 5.96 | 94.0 |
| ACGP | 2.408 | 9.05 | 0.9405 | 63.2 | 39.6 | 5.48 | 94.5 |
| Dynamic (4) | 1.358 | 4.80 | 0.9424 | 60.6 | 49.4 | 4.00 | 96.0 |
| Dynamic ($\infty$) | 0.615 | 1.65 | 0.9440 | 84.9 | 72.9 | 1.07 | 98.9 |

**Table 2.58:** QPE (exact), XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

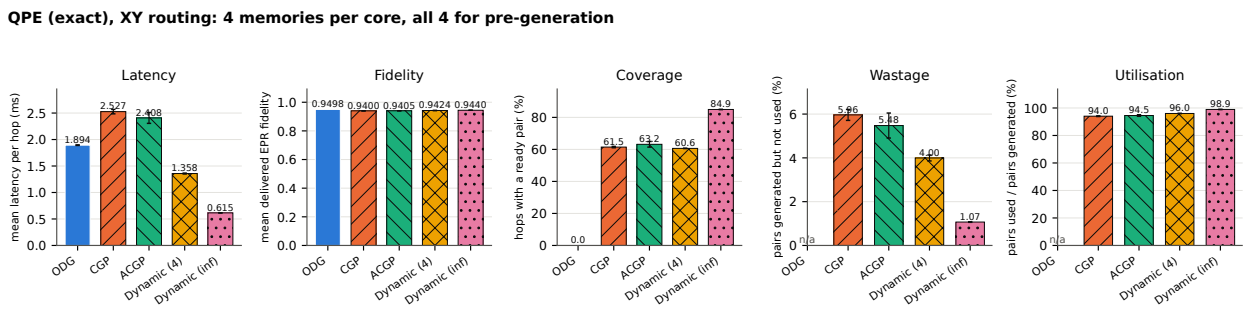



**Figure 2.58:** QPE (exact), XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.9.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 6.455 | 2.860 | 2.122 | 1.894 | 1.792 | 1.784 | 1.784 |
| CGP | 6.455 | 4.218 | 3.079 | 2.527 | 1.205 | 1.020 | 0.458 |
| ACGP | 6.455 | 4.345 | 3.096 | 2.408 | 1.223 | 1.016 | 0.457 |
| Dynamic | 8.929 | 3.513 | 1.897 | 1.358 | 0.897 | 0.758 | 0.615 |

**Table 2.59:** QPE (exact), XY routing: latency (mean latency per hop (ms)) for every K.

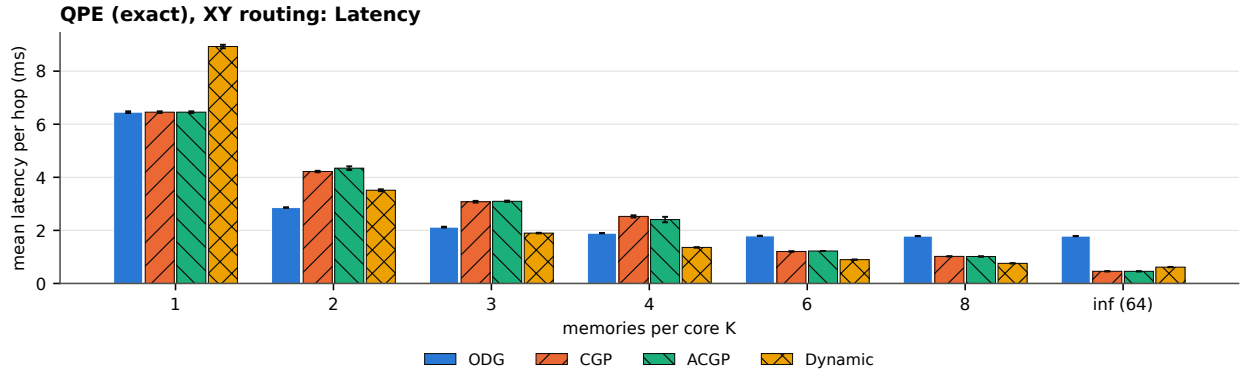



**Figure 2.59:** QPE (exact), XY routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9445 | 0.9419 | 0.9400 | 0.9435 | 0.9420 | 0.9421 |
| ACGP | 0.9498 | 0.9445 | 0.9419 | 0.9405 | 0.9431 | 0.9422 | 0.9423 |
| Dynamic | 0.9488 | 0.9444 | 0.9432 | 0.9424 | 0.9429 | 0.9435 | 0.9440 |

**Table 2.60:** QPE (exact), XY routing: fidelity (mean delivered EPR fidelity) for every K.

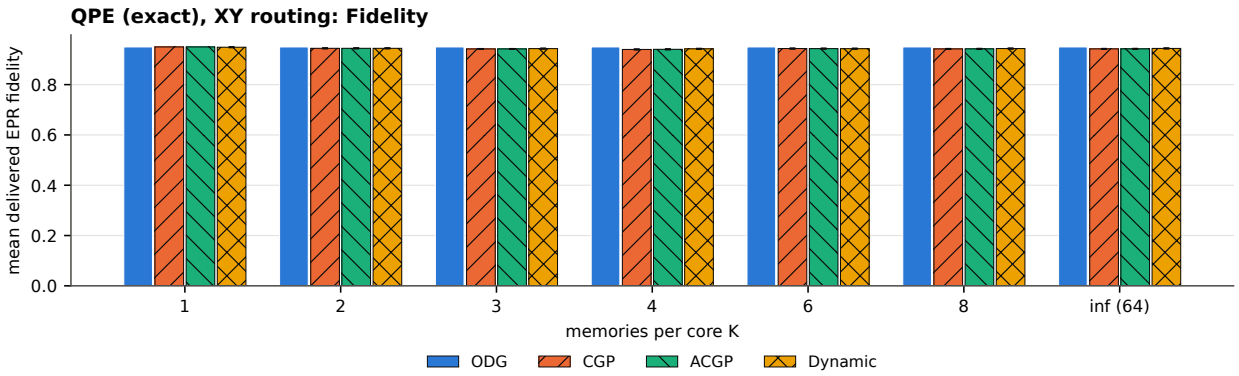



**Figure 2.60:** QPE (exact), XY routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 26.4 | 47.6 | 61.5 | 64.1 | 75.4 | 99.2 |
| ACGP | 0.0 | 26.1 | 47.1 | 63.2 | 64.0 | 76.1 | 99.1 |
| Dynamic | 10.0 | 32.8 | 48.4 | 60.6 | 76.7 | 77.0 | 84.9 |

**Table 2.61:** QPE (exact), XY routing: coverage (hops with a ready pair (%)) for every K.

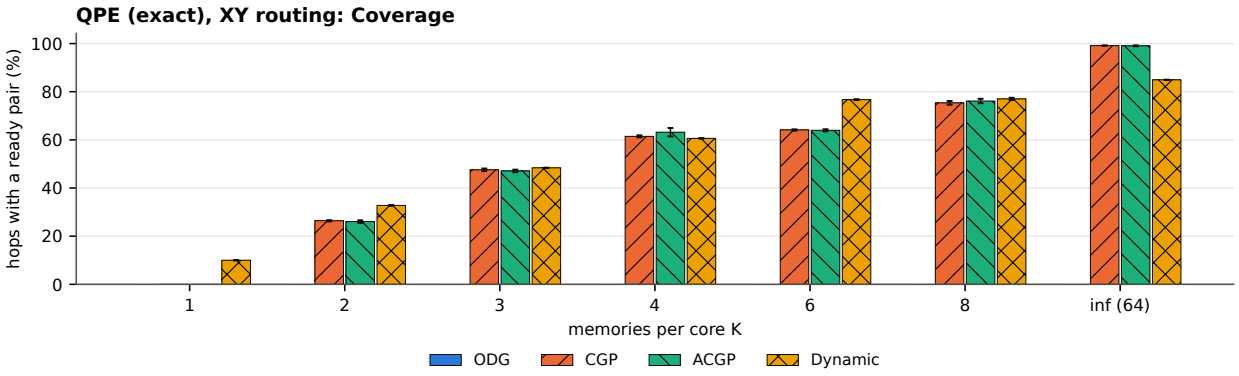



**Figure 2.61:** QPE (exact), XY routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 6.46 | 6.04 | 5.96 | 3.60 | 4.42 | 12.61 |
| ACGP | – | 6.71 | 5.66 | 5.48 | 3.78 | 4.32 | 12.50 |
| Dynamic | 49.72 | 13.88 | 6.38 | 4.00 | 2.09 | 0.99 | 1.07 |

**Table 2.62:** QPE (exact), XY routing: wastage (pairs generated but not used (%)) for every K.

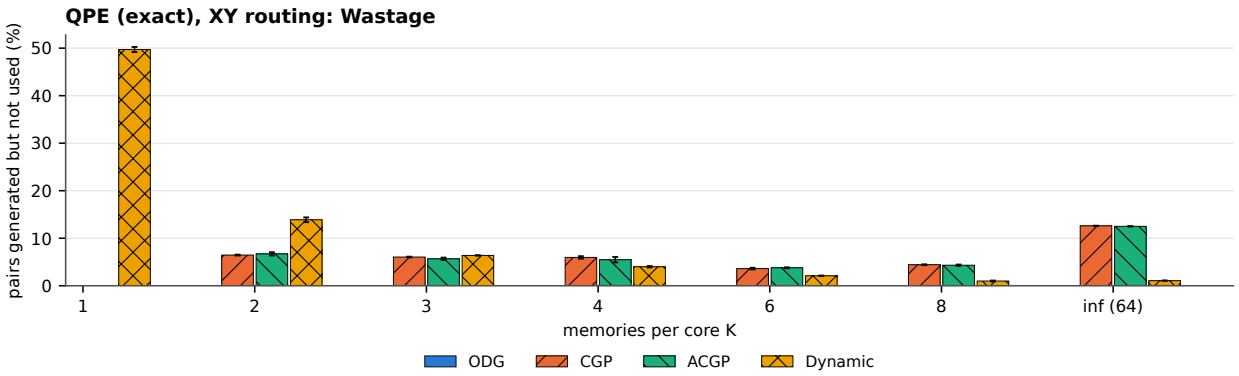



**Figure 2.62:** QPE (exact), XY routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 93.5 | 94.0 | 94.0 | 96.4 | 95.6 | 87.4 |
| ACGP | – | 93.3 | 94.3 | 94.5 | 96.2 | 95.7 | 87.5 |
| Dynamic | 50.3 | 86.1 | 93.6 | 96.0 | 97.9 | 99.0 | 98.9 |

**Table 2.63:** QPE (exact), XY routing: utilisation (pairs used / pairs generated (%)) for every K.

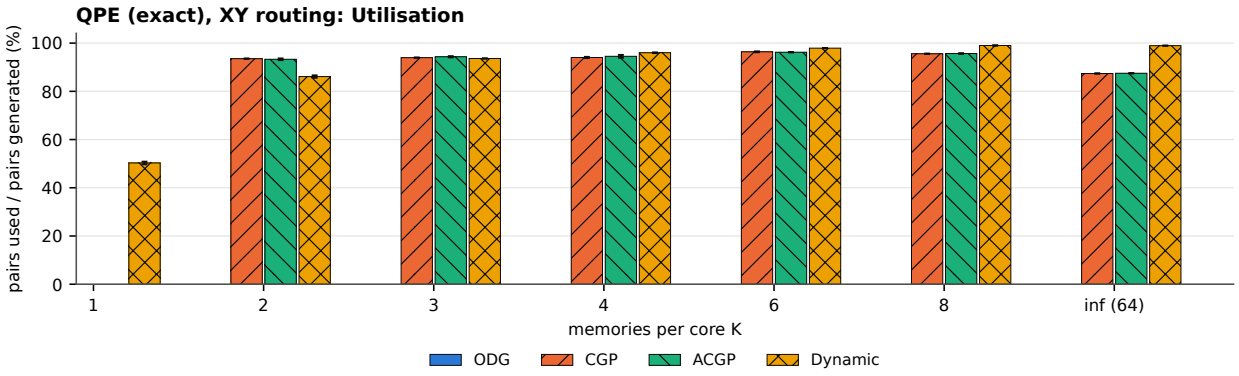



**Figure 2.63:** QPE (exact), XY routing: utilisation for every K.

## 2.10 QPE (exact), history routing

11921 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.10.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.774 | 4.40 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.947 | 2.95 | 0.9428 | 76.3 | 55.6 | 4.16 | 95.8 |
| ACGP | 0.960 | 3.10 | 0.9425 | 76.2 | 55.0 | 3.97 | 96.0 |
| Dynamic (8) | 0.714 | 2.20 | 0.9434 | 79.5 | 67.0 | 1.10 | 98.9 |
| Dynamic ($\infty$) | 0.578 | 1.40 | 0.9446 | 86.7 | 74.4 | 1.19 | 98.8 |

**Table 2.64:** QPE (exact), history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

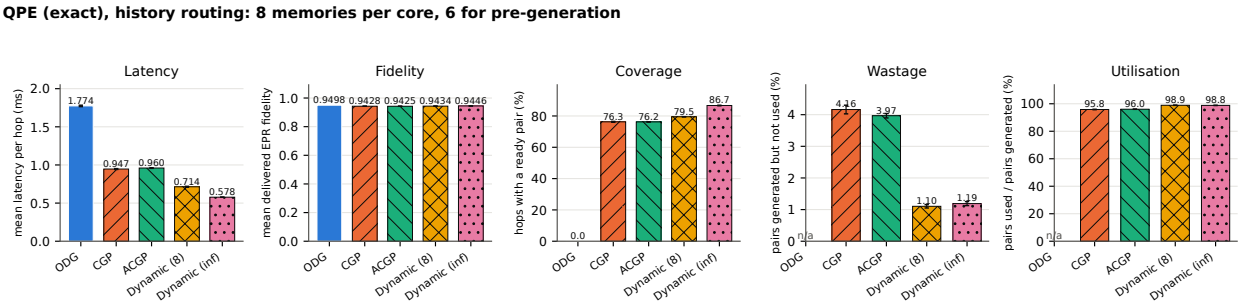



**Figure 2.64:** QPE (exact), history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.10.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.830 | 4.65 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 2.256 | 8.20 | 0.9416 | 61.6 | 36.2 | 5.21 | 94.8 |
| ACGP | 2.248 | 8.31 | 0.9413 | 63.4 | 38.6 | 5.33 | 94.7 |
| Dynamic (4) | 1.305 | 4.60 | 0.9417 | 62.6 | 51.4 | 4.57 | 95.4 |
| Dynamic ($\infty$) | 0.578 | 1.40 | 0.9446 | 86.7 | 74.4 | 1.19 | 98.8 |

**Table 2.65:** QPE (exact), history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

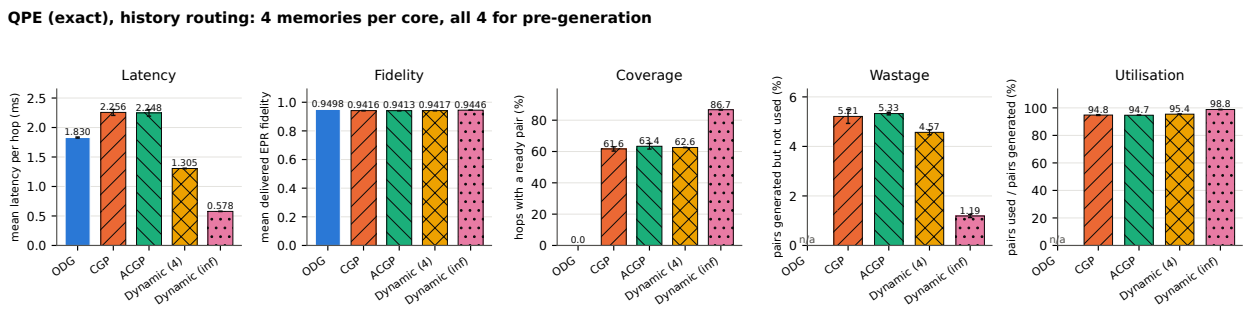



**Figure 2.65:** QPE (exact), history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.10.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 6.043 | 2.654 | 2.011 | 1.830 | 1.776 | 1.774 | 1.775 |
| CGP | 6.043 | 3.831 | 2.846 | 2.256 | 1.162 | 0.947 | 0.456 |
| ACGP | 6.043 | 3.843 | 2.809 | 2.248 | 1.127 | 0.960 | 0.456 |
| Dynamic | 8.359 | 3.260 | 1.832 | 1.305 | 0.856 | 0.714 | 0.578 |

**Table 2.66:** QPE (exact), history routing: latency (mean latency per hop (ms)) for every K.

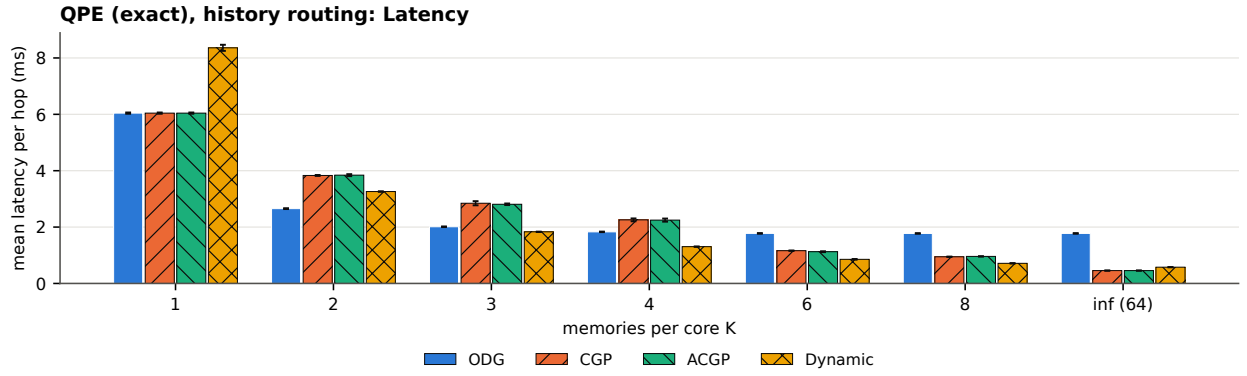



**Figure 2.66:** QPE (exact), history routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9451 | 0.9426 | 0.9416 | 0.9434 | 0.9428 | 0.9421 |
| ACGP | 0.9498 | 0.9450 | 0.9426 | 0.9413 | 0.9435 | 0.9425 | 0.9419 |
| Dynamic | 0.9485 | 0.9446 | 0.9426 | 0.9417 | 0.9428 | 0.9434 | 0.9446 |

**Table 2.67:** QPE (exact), history routing: fidelity (mean delivered EPR fidelity) for every K.

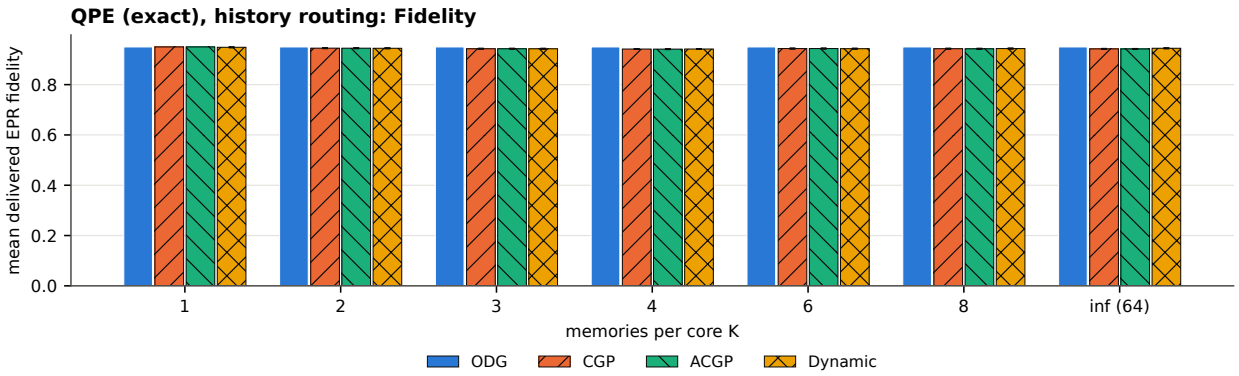



**Figure 2.67:** QPE (exact), history routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 26.7 | 47.6 | 61.6 | 63.6 | 76.3 | 99.3 |
| ACGP | 0.0 | 27.9 | 48.4 | 63.4 | 65.1 | 76.2 | 99.3 |
| Dynamic | 11.3 | 33.1 | 49.1 | 62.6 | 78.8 | 79.5 | 86.7 |

**Table 2.68:** QPE (exact), history routing: coverage (hops with a ready pair (%)) for every K.

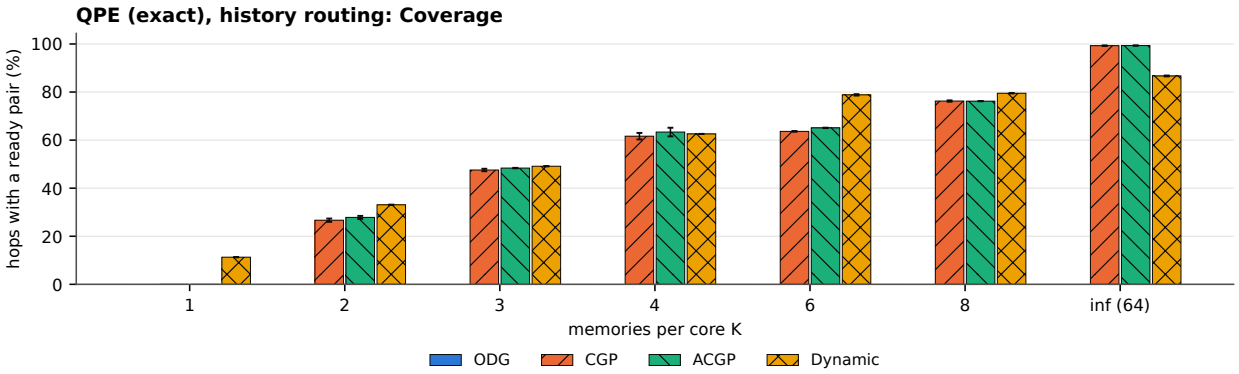



**Figure 2.68:** QPE (exact), history routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 5.93 | 5.44 | 5.21 | 3.71 | 4.16 | 12.64 |
| ACGP | – | 5.61 | 5.27 | 5.33 | 3.54 | 3.97 | 12.64 |
| Dynamic | 45.07 | 15.64 | 7.46 | 4.57 | 2.19 | 1.10 | 1.19 |

**Table 2.69:** QPE (exact), history routing: wastage (pairs generated but not used (%)) for every K.

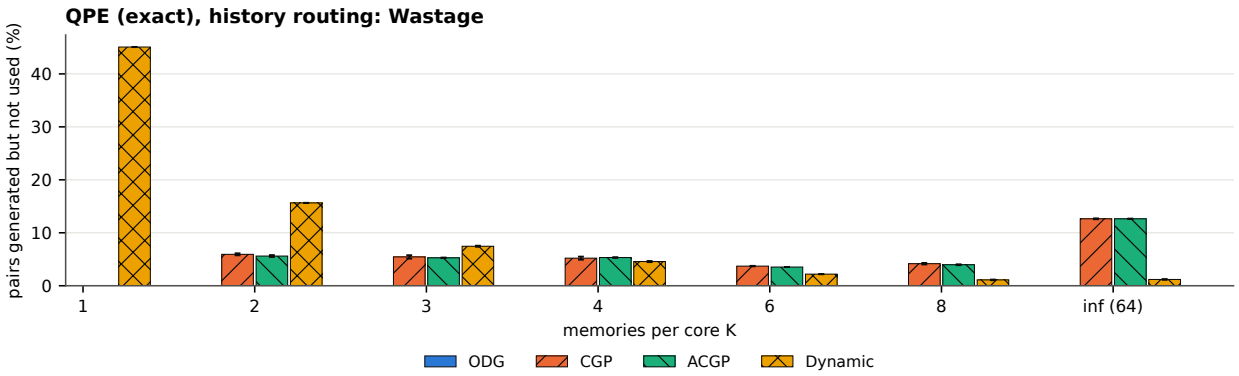



**Figure 2.69:** QPE (exact), history routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 94.1 | 94.6 | 94.8 | 96.3 | 95.8 | 87.4 |
| ACGP | – | 94.4 | 94.7 | 94.7 | 96.5 | 96.0 | 87.4 |
| Dynamic | 54.9 | 84.4 | 92.5 | 95.4 | 97.8 | 98.9 | 98.8 |

**Table 2.70:** QPE (exact), history routing: utilisation (pairs used / pairs generated (%)) for every K.

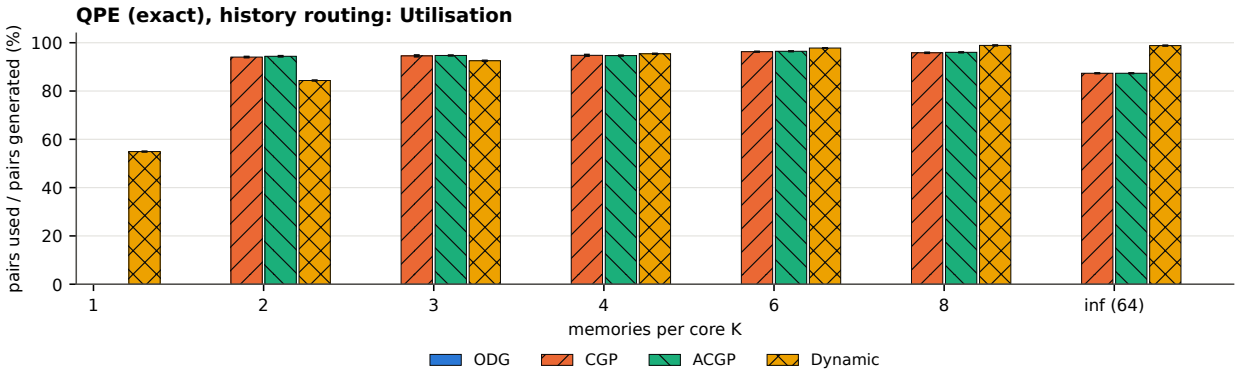



**Figure 2.70:** QPE (exact), history routing: utilisation for every K.

## 2.11 QAOA, XY routing

29605 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.11.1 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 8.860 | 3.673 | 2.493 | 2.076 | – | – | – |
| CGP | 8.860 | 5.593 | 3.951 | 2.984 | – | – | – |
| ACGP | 8.860 | 5.613 | 3.924 | 2.998 | – | – | – |
| Dynamic | 11.684 | 3.773 | 2.129 | – | – | – | – |

**Table 2.71:** QAOA, XY routing: latency (mean latency per hop (ms)) for every K.

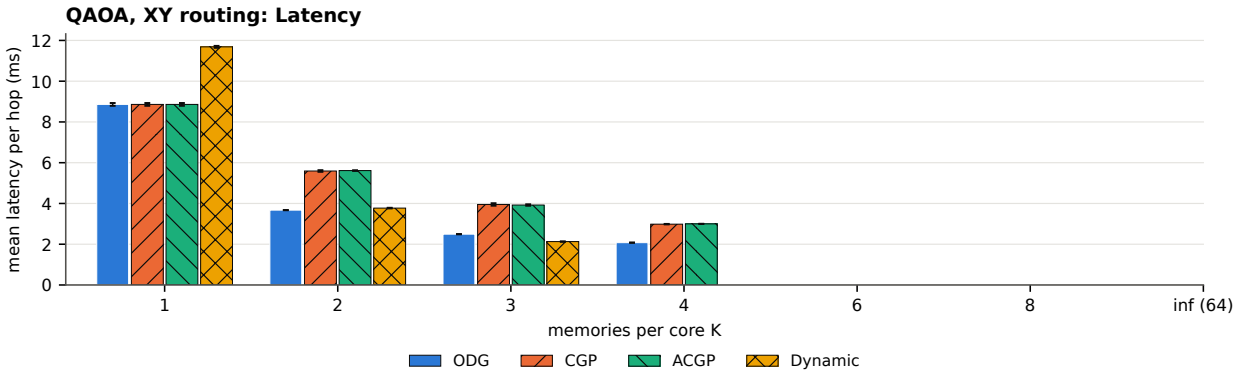



**Figure 2.71:** QAOA, XY routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | – | – | – |
| CGP | 0.9498 | 0.9447 | 0.9416 | 0.9400 | – | – | – |
| ACGP | 0.9498 | 0.9447 | 0.9416 | 0.9398 | – | – | – |
| Dynamic | 0.9487 | 0.9462 | 0.9449 | – | – | – | – |

**Table 2.72:** QAOA, XY routing: fidelity (mean delivered EPR fidelity) for every K.

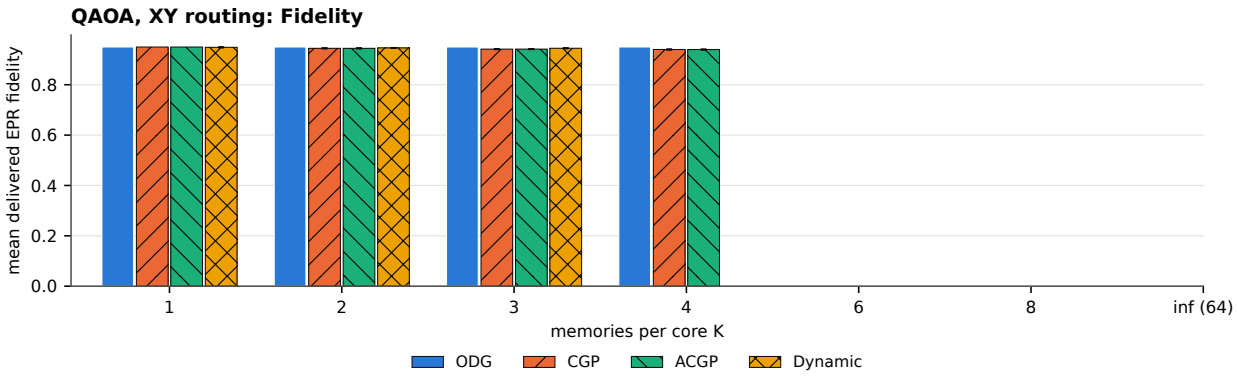



**Figure 2.72:** QAOA, XY routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | – | – | – |
| CGP | 0.0 | 26.9 | 48.8 | 63.5 | – | – | – |
| ACGP | 0.0 | 27.5 | 48.4 | 63.3 | – | – | – |
| Dynamic | 9.8 | 29.3 | 43.0 | – | – | – | – |

**Table 2.73:** QAOA, XY routing: coverage (hops with a ready pair (%)) for every K.

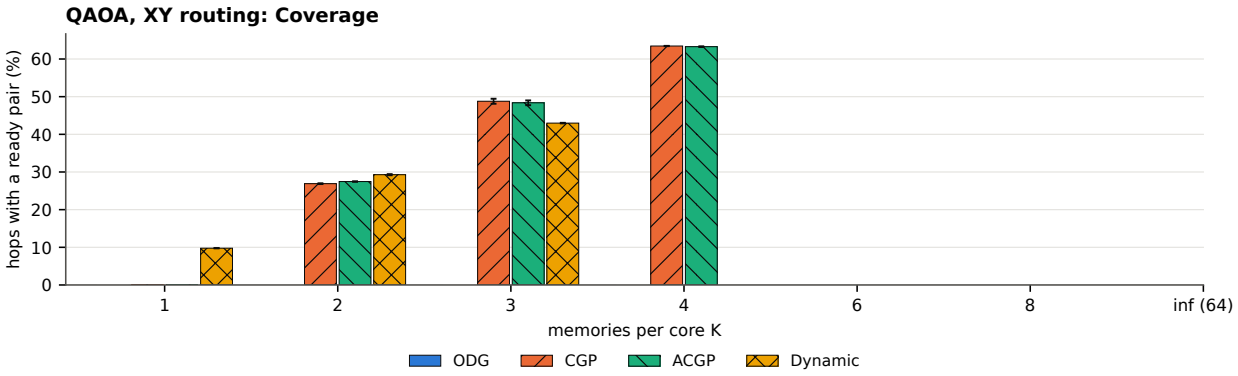



**Figure 2.73:** QAOA, XY routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 5.39 | 4.93 | 4.45 | – | – | – |
| ACGP | – | 5.51 | 4.71 | 4.61 | – | – | – |
| Dynamic | 37.78 | 8.38 | 4.63 | – | – | – | – |

**Table 2.74:** QAOA, XY routing: wastage (pairs generated but not used (%)) for every K.

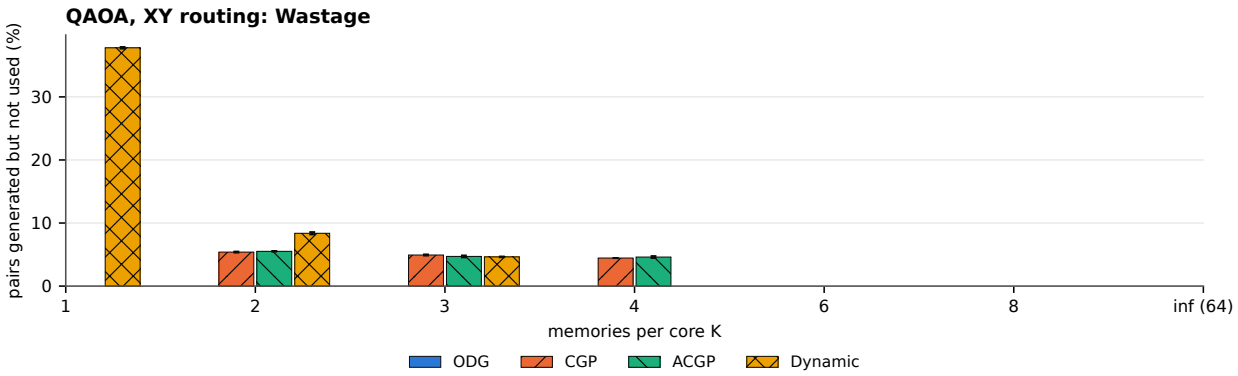



**Figure 2.74:** QAOA, XY routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 94.6 | 95.1 | 95.5 | – | – | – |
| ACGP | – | 94.5 | 95.3 | 95.4 | – | – | – |
| Dynamic | 62.2 | 91.6 | 95.4 | – | – | – | – |

**Table 2.75:** QAOA, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

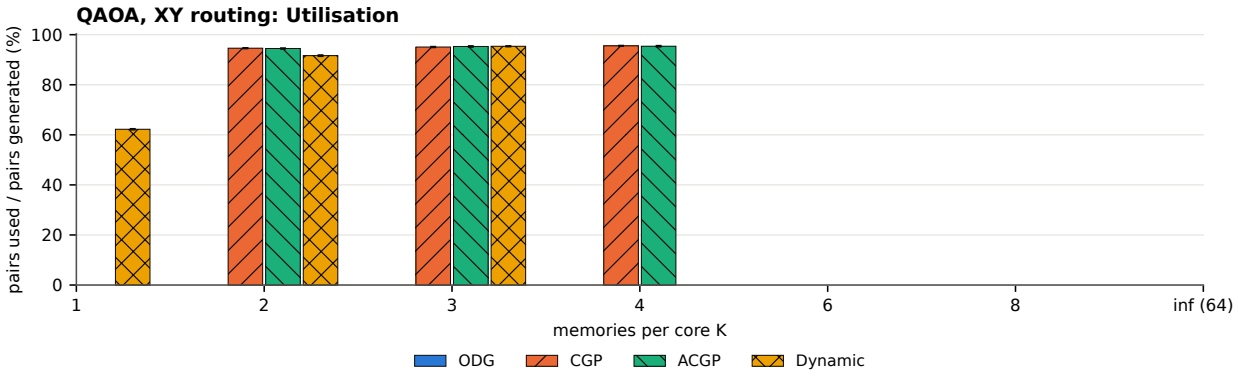



**Figure 2.75:** QAOA, XY routing: utilisation for every K.

## 2.12 QAOA, history routing

29605 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.12.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.819 | 4.65 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 1.170 | 3.65 | 0.9422 | 74.1 | 57.1 | 3.06 | 96.9 |
| ACGP | 1.170 | 3.65 | 0.9421 | 74.5 | 57.7 | 3.07 | 96.9 |
| Dynamic (8) | 0.846 | 2.70 | 0.9444 | 71.1 | 60.8 | 1.50 | 98.5 |
| Dynamic ($\infty$) | 0.609 | 1.60 | 0.9452 | 83.3 | 72.7 | 1.63 | 98.4 |

**Table 2.76:** QAOA, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

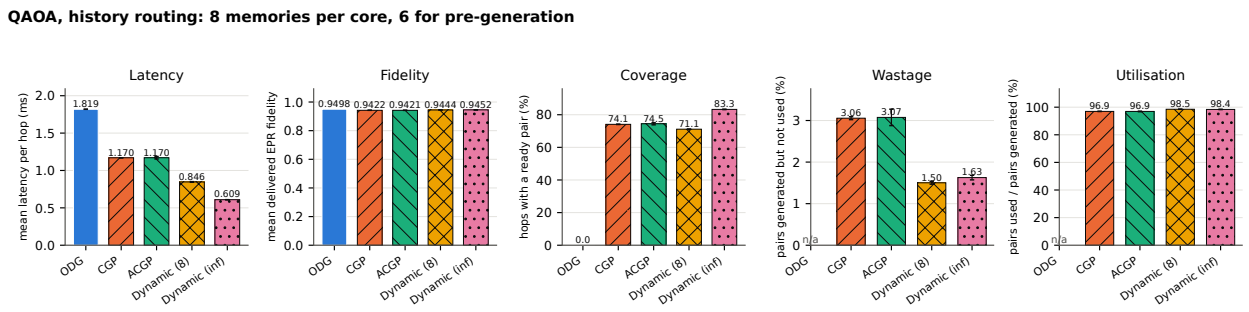



**Figure 2.76:** QAOA, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.12.2 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 8.382 | 3.463 | 2.378 | 2.019 | 1.841 | 1.819 | 1.818 |
| CGP | 8.382 | 5.103 | 3.671 | 2.787 | 1.438 | 1.170 | 0.458 |
| ACGP | 8.382 | 5.163 | 3.595 | 2.802 | 1.402 | 1.170 | 0.457 |
| Dynamic | 11.608 | 3.533 | 1.969 | 1.455 | 0.977 | 0.846 | 0.609 |

**Table 2.77:** QAOA, history routing: latency (mean latency per hop (ms)) for every K.

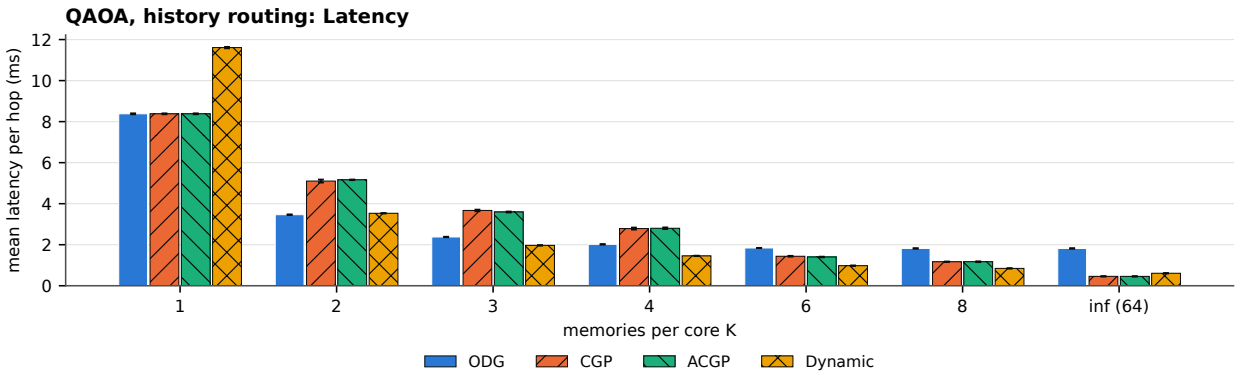



**Figure 2.77:** QAOA, history routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9451 | 0.9424 | 0.9409 | 0.9434 | 0.9422 | 0.9418 |
| ACGP | 0.9498 | 0.9450 | 0.9425 | 0.9409 | 0.9436 | 0.9421 | 0.9417 |
| Dynamic | 0.9486 | 0.9460 | 0.9448 | 0.9438 | 0.9439 | 0.9444 | 0.9452 |

**Table 2.78:** QAOA, history routing: fidelity (mean delivered EPR fidelity) for every K.

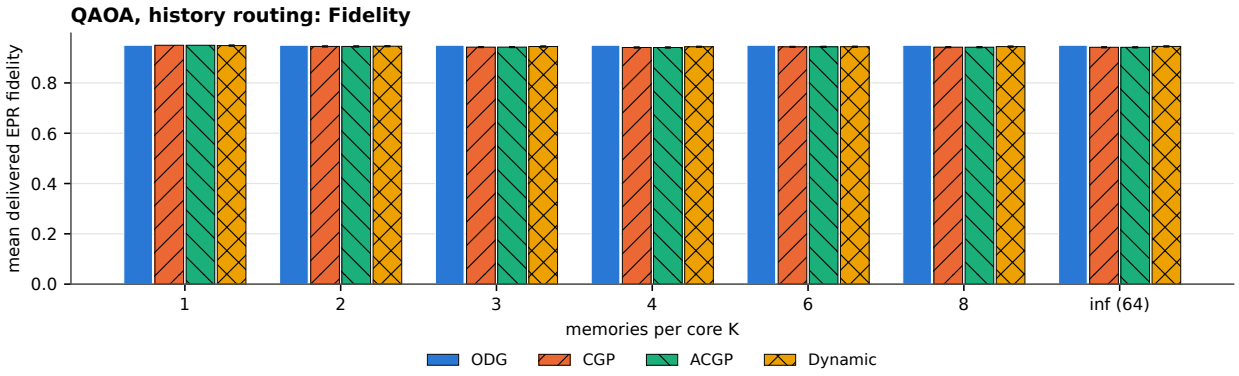



**Figure 2.78:** QAOA, history routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 28.0 | 48.8 | 63.4 | 60.6 | 74.1 | 99.1 |
| ACGP | 0.0 | 28.5 | 49.5 | 63.4 | 61.9 | 74.5 | 99.2 |
| Dynamic | 10.1 | 30.2 | 44.3 | 56.3 | 72.0 | 71.1 | 83.3 |

**Table 2.79:** QAOA, history routing: coverage (hops with a ready pair (%)) for every K.

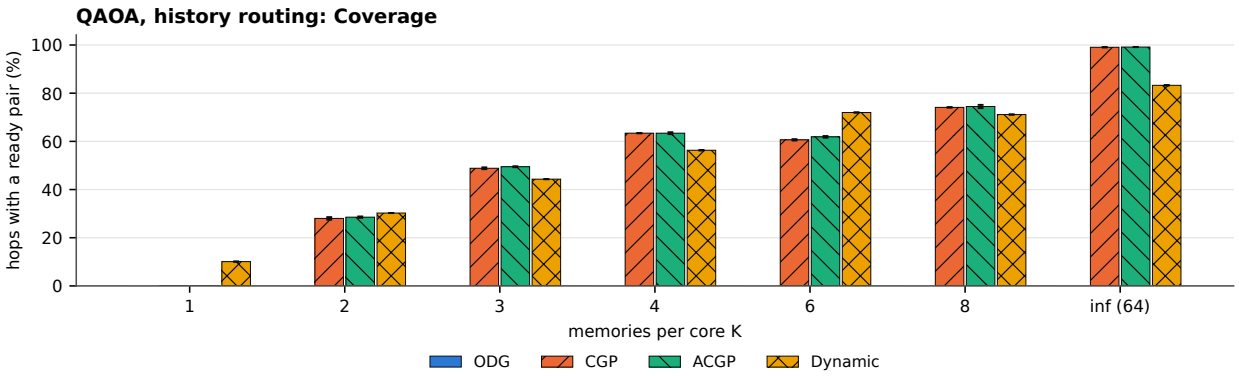



**Figure 2.79:** QAOA, history routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 4.78 | 4.31 | 4.02 | 2.85 | 3.06 | 8.36 |
| ACGP | – | 4.73 | 4.21 | 3.85 | 2.67 | 3.07 | 8.36 |
| Dynamic | 37.35 | 8.46 | 4.74 | 3.27 | 1.98 | 1.50 | 1.63 |

**Table 2.80:** QAOA, history routing: wastage (pairs generated but not used (%)) for every K.

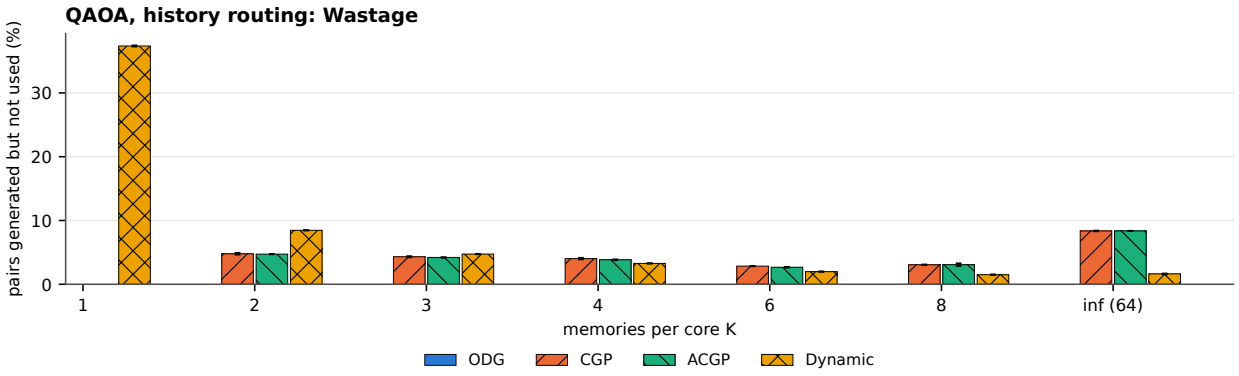



**Figure 2.80:** QAOA, history routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 95.2 | 95.7 | 96.0 | 97.2 | 96.9 | 91.6 |
| ACGP | – | 95.3 | 95.8 | 96.2 | 97.3 | 96.9 | 91.6 |
| Dynamic | 62.6 | 91.5 | 95.3 | 96.7 | 98.0 | 98.5 | 98.4 |

**Table 2.81:** QAOA, history routing: utilisation (pairs used / pairs generated (%)) for every K.

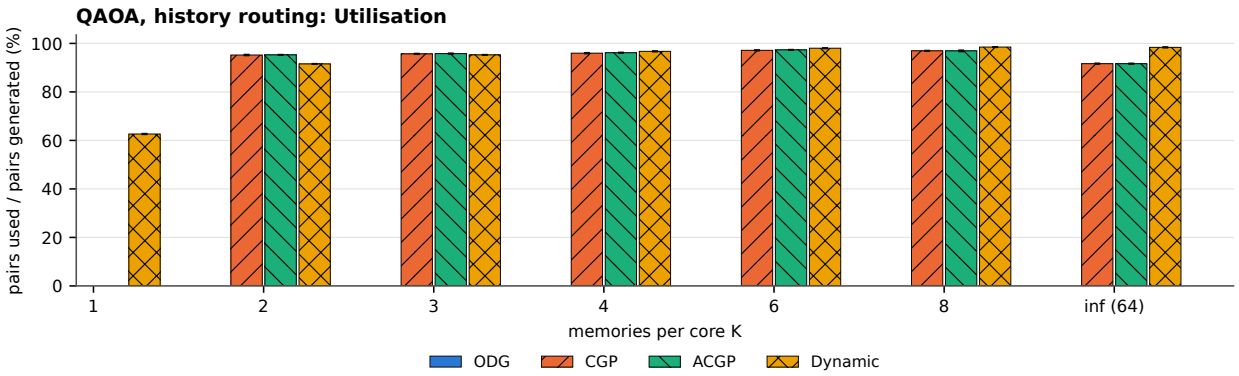



**Figure 2.81:** QAOA, history routing: utilisation for every K.

## 2.13 Draper adder, XY routing

6510 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.13.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.761 | 4.20 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.687 | 1.90 | 0.9404 | 81.5 | 68.7 | 6.00 | 94.0 |
| ACGP | 0.671 | 1.80 | 0.9405 | 81.7 | 69.6 | 6.03 | 94.0 |
| Dynamic (8) | 0.623 | 1.60 | 0.9467 | 79.2 | 65.8 | 2.43 | 97.6 |
| Dynamic ($\infty$) | 0.611 | 1.50 | 0.9467 | 79.9 | 66.6 | 2.74 | 97.3 |

**Table 2.82:** Draper adder, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

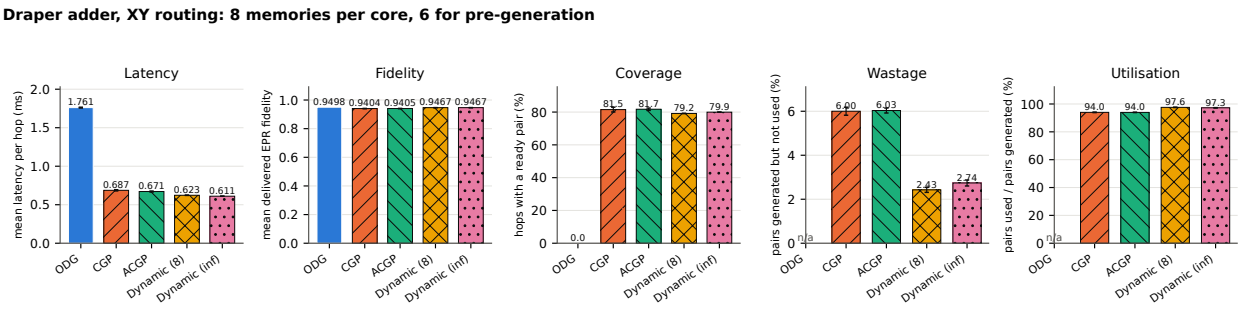



**Figure 2.82:** Draper adder, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.13.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.765 | 4.20 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 1.351 | 4.50 | 0.9400 | 65.5 | 46.6 | 6.00 | 94.0 |
| ACGP | 1.347 | 4.50 | 0.9395 | 65.8 | 47.7 | 6.14 | 93.9 |
| Dynamic (4) | 0.783 | 2.35 | 0.9465 | 77.6 | 64.0 | 3.00 | 97.0 |
| Dynamic ($\infty$) | 0.611 | 1.50 | 0.9467 | 79.9 | 66.6 | 2.74 | 97.3 |

**Table 2.83:** Draper adder, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

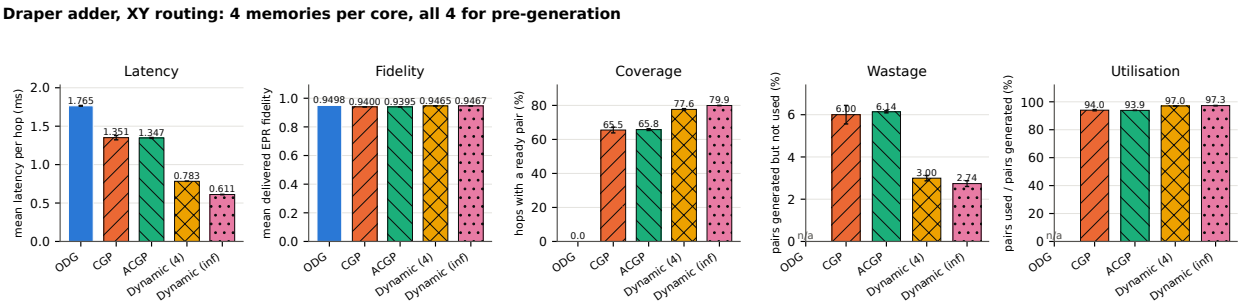



**Figure 2.83:** Draper adder, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.13.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 3.506 | 1.918 | 1.783 | 1.765 | 1.761 | 1.761 | 1.761 |
| CGP | 3.506 | 2.263 | 1.699 | 1.351 | 0.826 | 0.687 | 0.456 |
| ACGP | 3.506 | 2.271 | 1.668 | 1.347 | 0.784 | 0.671 | 0.455 |
| Dynamic | 3.975 | 1.546 | 1.014 | 0.783 | 0.645 | 0.623 | 0.611 |

**Table 2.84:** Draper adder, XY routing: latency (mean latency per hop (ms)) for every K.

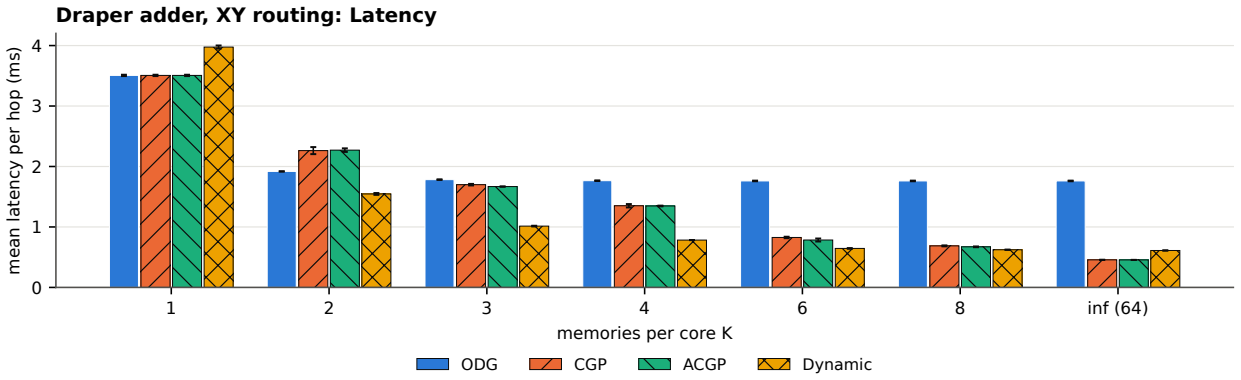



**Figure 2.84:** Draper adder, XY routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9441 | 0.9413 | 0.9400 | 0.9410 | 0.9404 | 0.9405 |
| ACGP | 0.9498 | 0.9441 | 0.9412 | 0.9395 | 0.9410 | 0.9405 | 0.9405 |
| Dynamic | 0.9480 | 0.9454 | 0.9459 | 0.9465 | 0.9467 | 0.9467 | 0.9467 |

**Table 2.85:** Draper adder, XY routing: fidelity (mean delivered EPR fidelity) for every K.

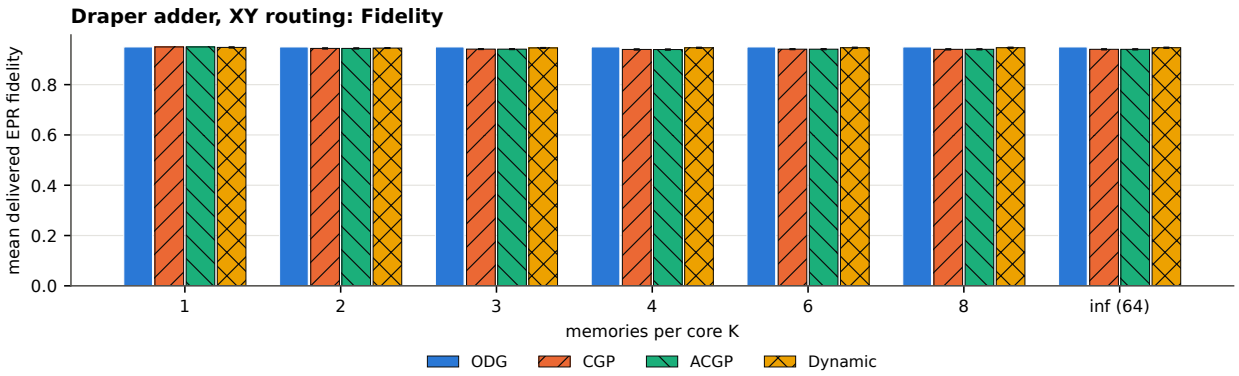



**Figure 2.85:** Draper adder, XY routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 29.2 | 49.7 | 65.5 | 70.9 | 81.5 | 99.2 |
| ACGP | 0.0 | 28.4 | 51.6 | 65.8 | 73.8 | 81.7 | 99.5 |
| Dynamic | 21.0 | 58.1 | 71.3 | 77.6 | 79.7 | 79.2 | 79.9 |

**Table 2.86:** Draper adder, XY routing: coverage (hops with a ready pair (%)) for every K.

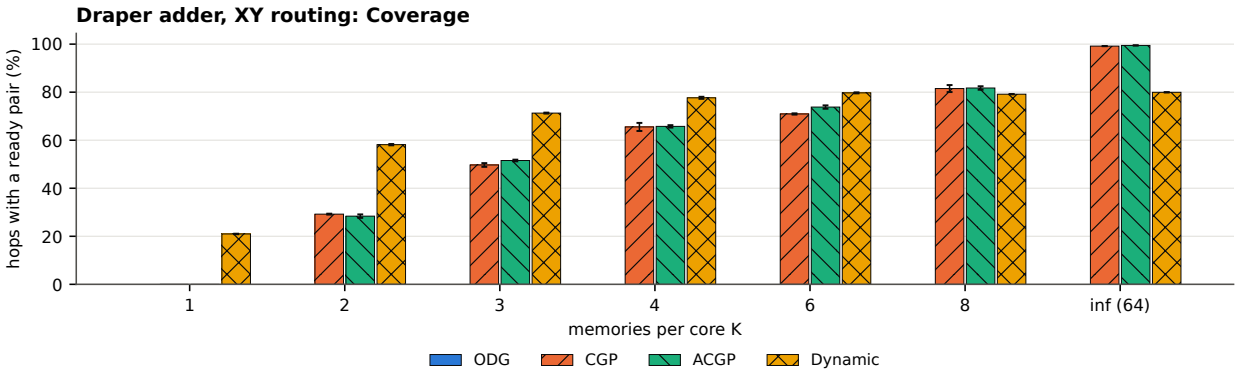



**Figure 2.86:** Draper adder, XY routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 7.18 | 7.07 | 6.00 | 5.66 | 6.00 | 23.84 |
| ACGP | – | 7.39 | 6.42 | 6.14 | 5.16 | 6.03 | 24.02 |
| Dynamic | 41.98 | 9.78 | 3.73 | 3.00 | 2.68 | 2.43 | 2.74 |

**Table 2.87:** Draper adder, XY routing: wastage (pairs generated but not used (%)) for every K.

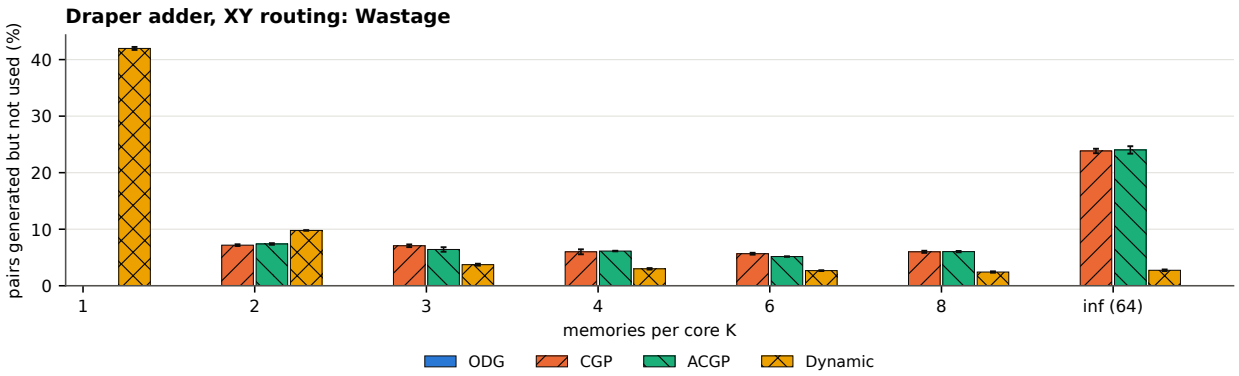



**Figure 2.87:** Draper adder, XY routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 92.8 | 92.9 | 94.0 | 94.3 | 94.0 | 76.2 |
| ACGP | – | 92.6 | 93.6 | 93.9 | 94.8 | 94.0 | 76.0 |
| Dynamic | 58.0 | 90.2 | 96.3 | 97.0 | 97.3 | 97.6 | 97.3 |

**Table 2.88:** Draper adder, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

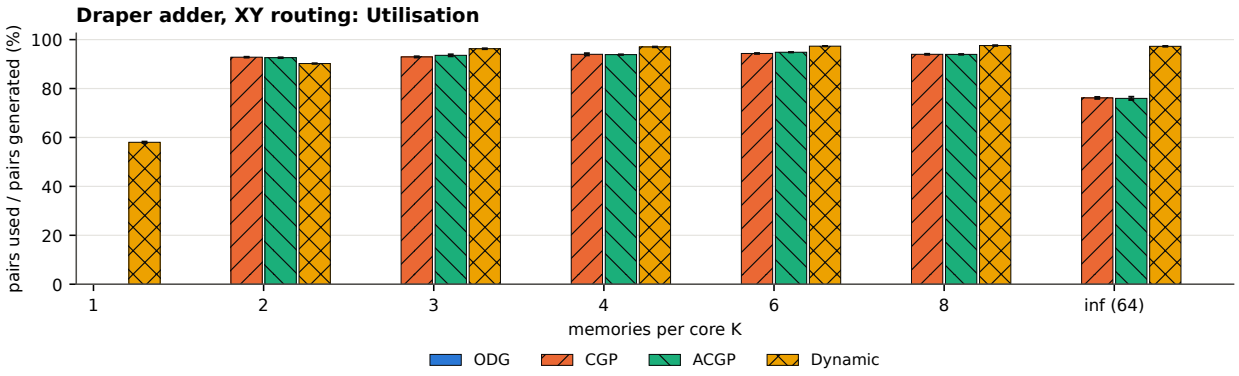



**Figure 2.88:** Draper adder, XY routing: utilisation for every K.

## 2.14 Draper adder, history routing

6510 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

### 2.14.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic ($\infty$) uses 64, which no core ever fills.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.765 | 4.40 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 0.689 | 1.90 | 0.9410 | 80.6 | 67.8 | 6.24 | 93.8 |
| ACGP | 0.689 | 1.99 | 0.9410 | 81.1 | 68.1 | 6.14 | 93.9 |
| Dynamic (8) | 0.628 | 1.60 | 0.9467 | 78.5 | 64.9 | 2.58 | 97.4 |
| Dynamic ($\infty$) | 0.611 | 1.50 | 0.9466 | 80.0 | 66.9 | 2.45 | 97.5 |

**Table 2.89:** Draper adder, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

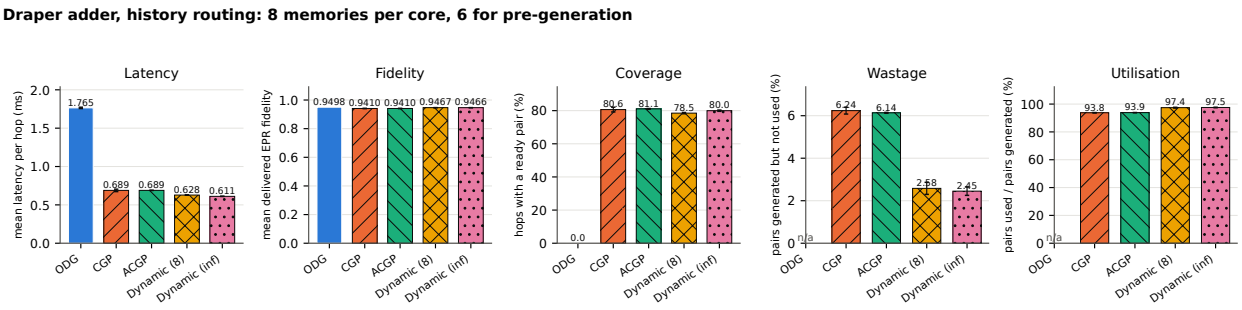



**Figure 2.89:** Draper adder, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.14.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the $K$ sweep; Dynamic ($\infty$) as above.

| Configuration | Latency (ms) | p95 (ms) | Fidelity | Hop cov. (%) | Chain full (%) | Wastage (%) | Utilisation (%) |
|---|---|---|---|---|---|---|---|
| ODG | 1.768 | 4.40 | 0.9498 | 0.0 | 0.0 | – | – |
| CGP | 1.352 | 4.45 | 0.9404 | 65.0 | 47.5 | 6.34 | 93.7 |
| ACGP | 1.292 | 4.15 | 0.9409 | 67.2 | 51.0 | 5.98 | 94.0 |
| Dynamic (4) | 0.788 | 2.40 | 0.9462 | 77.5 | 64.5 | 2.86 | 97.1 |
| Dynamic ($\infty$) | 0.611 | 1.50 | 0.9466 | 80.0 | 66.9 | 2.45 | 97.5 |

**Table 2.90:** Draper adder, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

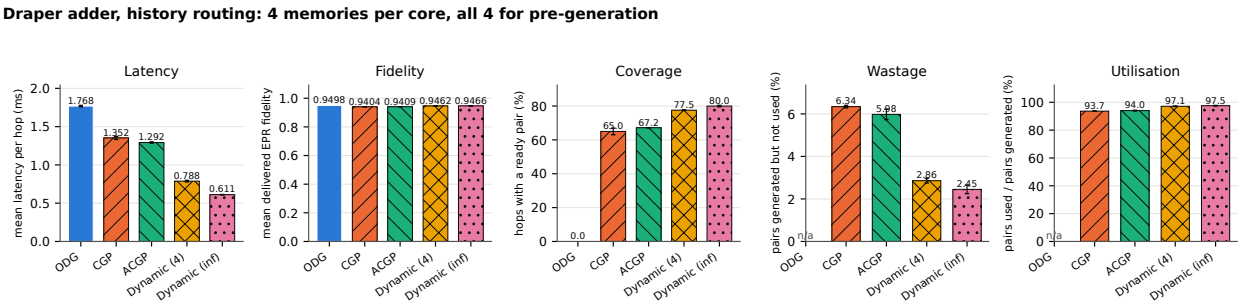



**Figure 2.90:** Draper adder, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

### 2.14.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

| Latency | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 3.445 | 1.929 | 1.789 | 1.768 | 1.765 | 1.765 | 1.765 |
| CGP | 3.445 | 2.233 | 1.675 | 1.352 | 0.823 | 0.689 | 0.455 |
| ACGP | 3.445 | 2.209 | 1.664 | 1.292 | 0.845 | 0.689 | 0.456 |
| Dynamic | 3.857 | 1.505 | 0.993 | 0.788 | 0.650 | 0.628 | 0.611 |

**Table 2.91:** Draper adder, history routing: latency (mean latency per hop (ms)) for every K.

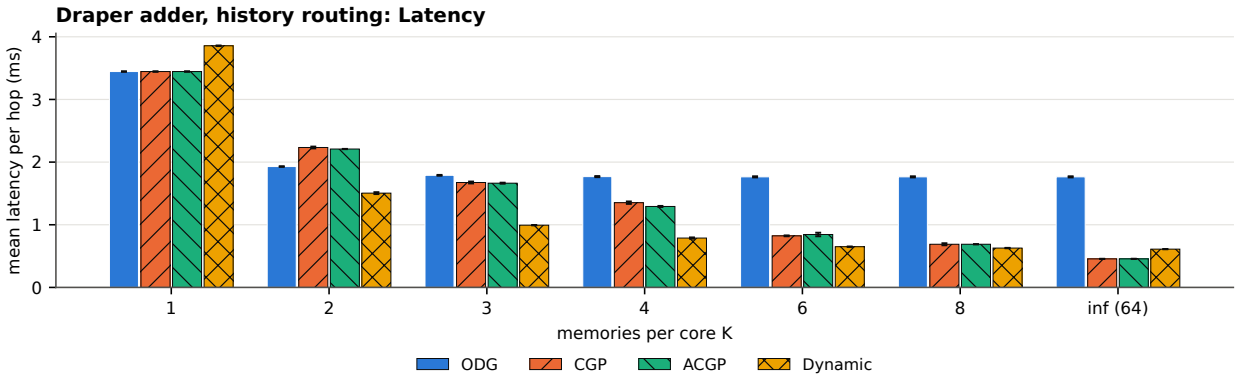



**Figure 2.91:** Draper adder, history routing: latency for every K.

| Fidelity | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 | 0.9498 |
| CGP | 0.9498 | 0.9442 | 0.9416 | 0.9404 | 0.9414 | 0.9410 | 0.9407 |
| ACGP | 0.9498 | 0.9445 | 0.9418 | 0.9409 | 0.9410 | 0.9410 | 0.9405 |
| Dynamic | 0.9478 | 0.9452 | 0.9458 | 0.9462 | 0.9466 | 0.9467 | 0.9466 |

**Table 2.92:** Draper adder, history routing: fidelity (mean delivered EPR fidelity) for every K.

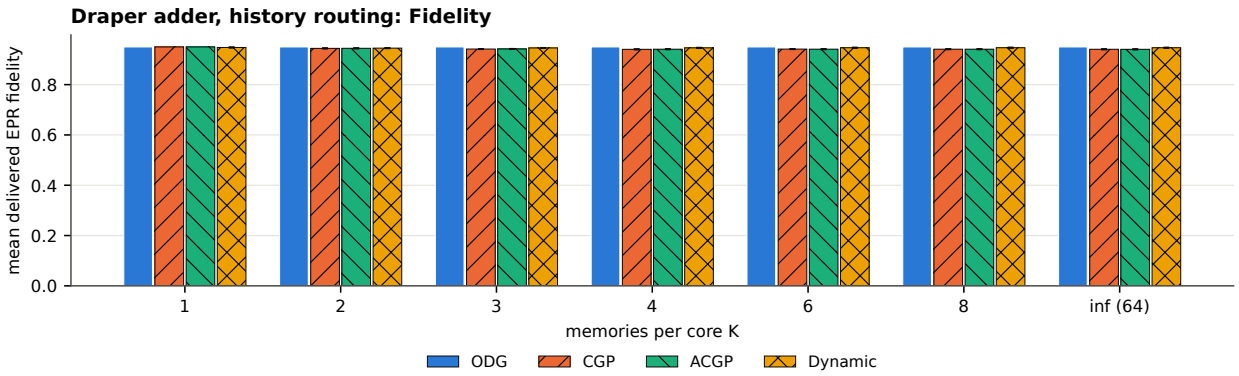



**Figure 2.92:** Draper adder, history routing: fidelity for every K.

| Coverage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| CGP | 0.0 | 30.7 | 50.5 | 65.0 | 71.2 | 80.6 | 99.4 |
| ACGP | 0.0 | 30.6 | 52.0 | 67.2 | 70.7 | 81.1 | 99.3 |
| Dynamic | 21.9 | 61.6 | 71.9 | 77.5 | 79.5 | 78.5 | 80.0 |

**Table 2.93:** Draper adder, history routing: coverage (hops with a ready pair (%)) for every K.

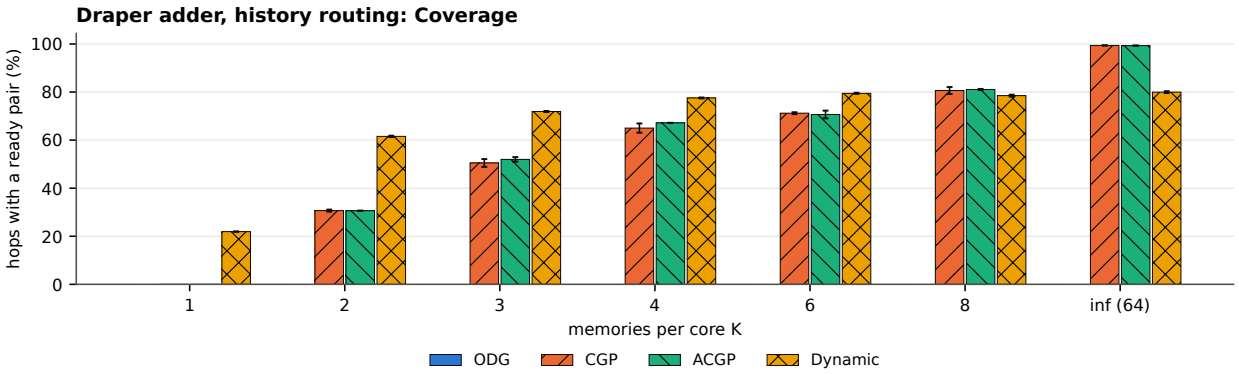



**Figure 2.93:** Draper adder, history routing: coverage for every K.

| Wastage | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 6.90 | 6.81 | 6.34 | 5.52 | 6.24 | 23.86 |
| ACGP | – | 6.56 | 6.56 | 5.98 | 5.96 | 6.14 | 24.23 |
| Dynamic | 42.34 | 9.89 | 4.14 | 2.86 | 2.60 | 2.58 | 2.45 |

**Table 2.94:** Draper adder, history routing: wastage (pairs generated but not used (%)) for every K.

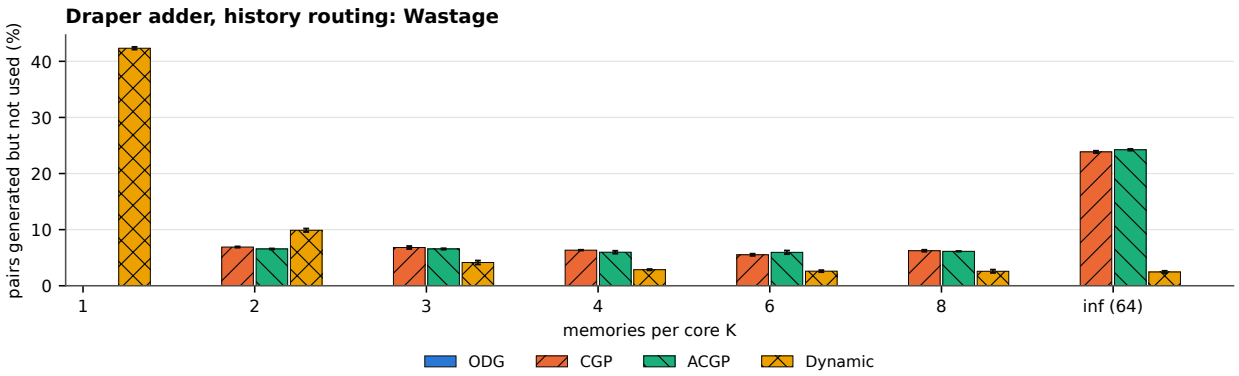



**Figure 2.94:** Draper adder, history routing: wastage for every K.

| Utilisation | $K = 1$ | $K = 2$ | $K = 3$ | $K = 4$ | $K = 6$ | $K = 8$ | $K = \infty$ |
|---|---|---|---|---|---|---|---|
| ODG | – | – | – | – | – | – | – |
| CGP | – | 93.1 | 93.2 | 93.7 | 94.5 | 93.8 | 76.1 |
| ACGP | – | 93.4 | 93.4 | 94.0 | 94.0 | 93.9 | 75.8 |
| Dynamic | 57.7 | 90.1 | 95.9 | 97.1 | 97.4 | 97.4 | 97.5 |

**Table 2.95:** Draper adder, history routing: utilisation (pairs used / pairs generated (%)) for every K.

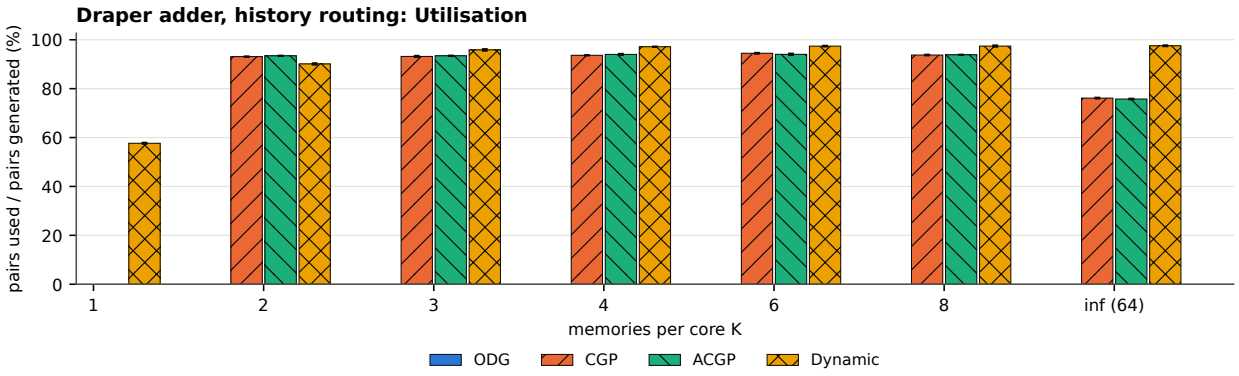



**Figure 2.95:** Draper adder, history routing: utilisation for every K.